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Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mathematics

Advanced Subsidiary
Paper 1: Pure Mathematics

Wednesday 16 May 2018 – Morning
Time: 2 hours

Paper Reference

8MA0/01

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. Find

$$\int \left(\frac{2}{3}x^3 - 6\sqrt{x} + 1 \right) dx$$

giving your answer in its simplest form.

(4)

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Question 1 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 1 is 4 marks)



2. (i) Show that $x^2 - 8x + 17 > 0$ for all real values of x

(3)

(ii) "If I add 3 to a number and square the sum, the result is greater than the square of the original number."

State, giving a reason, if the above statement is always true, sometimes true or never true.

(2)

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Question 2 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 2 is 5 marks)



3. Given that the point A has position vector $4\mathbf{i} - 5\mathbf{j}$ and the point B has position vector $-5\mathbf{i} - 2\mathbf{j}$,

(a) find the vector $\overrightarrow{AB}$,

(2)

(b) find $|\overrightarrow{AB}|$.

Give your answer as a simplified surd.

(2)

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Question 3 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 3 is 4 marks)



4. The line l_1 has equation $4y - 3x = 10$

The line l_2 passes through the points $(5, -1)$ and $(-1, 8)$.

Determine, giving full reasons for your answer, whether lines l_1 and l_2 are parallel, perpendicular or neither.

(4)

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 4 marks)



5. A student's attempt to solve the equation $2\log_2 x - \log_2 \sqrt{x} = 3$ is shown below.

$$2\log_2 x - \log_2 \sqrt{x} = 3$$

$$2\log_2 \left(\frac{x}{\sqrt{x}} \right) = 3$$

using the subtraction law for logs

$$2\log_2 (\sqrt{x}) = 3$$

simplifying

$$\log_2 x = 3$$

using the power law for logs

$$x = 3^2 = 9$$

using the definition of a log

(a) Identify two errors made by this student, giving a brief explanation of each.

(2)

(b) Write out the correct solution.

(3)

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



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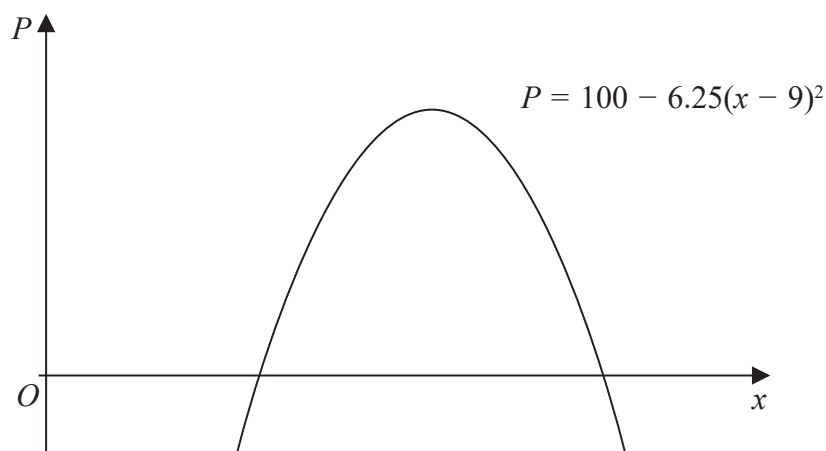


Figure 1

A company makes a particular type of children's toy.

The annual profit made by the company is modelled by the equation

$$P = 100 - 6.25(x - 9)^2$$

where P is the profit measured in thousands of pounds and x is the selling price of the toy in pounds.

A sketch of P against x is shown in Figure 1.

Using the model,

- (a) explain why £15 is not a sensible selling price for the toy. (2)

Given that the company made an annual profit of more than £80 000

- (b) find, according to the model, the least possible selling price for the toy. (3)

The company wishes to maximise its annual profit.

State, according to the model,

- (c) (i) the maximum possible annual profit,
(ii) the selling price of the toy that maximises the annual profit. (2)



Question 6 continued

Handwriting practice area with 20 horizontal lines.

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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 7 marks)



7. In a triangle ABC , side AB has length 10 cm, side AC has length 5 cm, and angle $BAC = \theta$ where θ is measured in degrees. The area of triangle ABC is 15 cm^2

(a) Find the two possible values of $\cos \theta$

(4)

Given that BC is the longest side of the triangle,

(b) find the exact length of BC .

(2)

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 6 marks)



8. A lorry is driven between London and Newcastle.

In a simple model, the cost of the journey £ C when the lorry is driven at a steady speed of v kilometres per hour is

$$C = \frac{1500}{v} + \frac{2v}{11} + 60$$

(a) Find, according to this model,

(i) the value of v that minimises the cost of the journey,

(ii) the minimum cost of the journey.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(6)

(b) Prove by using $\frac{d^2C}{dv^2}$ that the cost is minimised at the speed found in (a)(i).

(2)

(c) State one limitation of this model.

(1)

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Question 8 continued

Lined area for writing the answer to Question 8 continued.

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Question 8 continued

Lined area for writing the answer to Question 8.

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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 9 marks)



Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing the answer to Question 9.

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 9 marks)



10. Prove, from first principles, that the derivative of x^3 is $3x^2$

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 4 marks)



11. (a) Find the first 3 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{x}{16}\right)^9$$

giving each term in its simplest form.

(4)

$$f(x) = (a + bx)\left(2 - \frac{x}{16}\right)^9, \text{ where } a \text{ and } b \text{ are constants}$$

Given that the first two terms, in ascending powers of x , in the series expansion of $f(x)$ are 128 and $36x$,

(b) find the value of a ,

(2)

(c) find the value of b .

(2)

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Question 11 continued

Lined area for writing the answer to Question 11.

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Question 11 continued

Lined area for writing the answer to Question 11.

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Question 11 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 11 is 8 marks)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

Lined area for writing the answer to Question 12.

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 8 marks)



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Question 13 continued

Lined area for writing the answer to Question 13.

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Question 13 continued

Lined area for writing the answer to Question 13.

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 8 marks)



14. The circle C has equation

$$x^2 + y^2 - 6x + 10y + 9 = 0$$

(a) Find

(i) the coordinates of the centre of C

(ii) the radius of C

(3)

The line with equation $y = kx$, where k is a constant, cuts C at two distinct points.

(b) Find the range of values for k .

(6)

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Question 14 continued

Lined area for writing the answer to Question 14.

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Question 14 continued

Lined area for writing the answer to Question 14.

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 9 marks)



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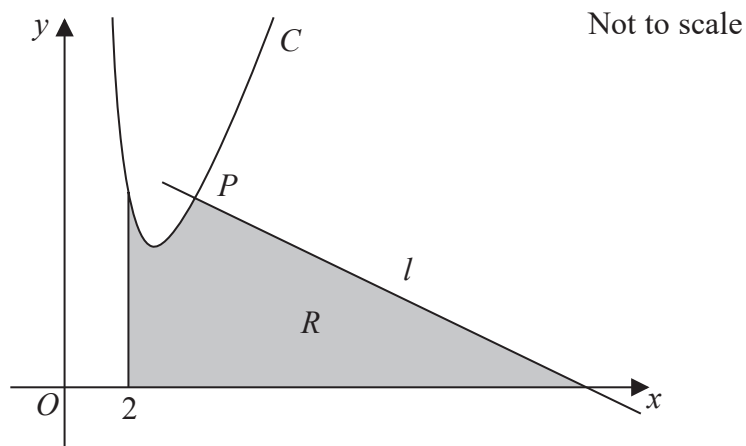
**Figure 4**

Figure 4 shows a sketch of part of the curve C with equation

$$y = \frac{32}{x^2} + 3x - 8, \quad x > 0$$

The point $P(4, 6)$ lies on C .

The line l is the normal to C at the point P .

The region R , shown shaded in Figure 4, is bounded by the line l , the curve C , the line with equation $x = 2$ and the x -axis.

Show that the area of R is 46

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(10)

Question 15 continued

Lined area for writing the answer to Question 15.

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Question 15 continued

Lined area for writing the answer to Question 15.

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Question 15 continued

Handwriting practice area with 20 horizontal lines.

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Question 15 continued

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(Total for Question 15 is 10 marks)

TOTAL FOR THE PAPER: 100 MARKS



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Wednesday 15 May 2019			
Morning (Time: 2 hours)		Paper Reference 8MA0/01	
Mathematics Advanced Subsidiary Paper 1: Pure Mathematics			
You must have: Mathematical Formulae and Statistical Tables, calculator			Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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Answers without working may not gain full credit.
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Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

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- Check your answers if you have time at the end.

Turn over ►

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1. The line l_1 has equation $2x + 4y - 3 = 0$

The line l_2 has equation $y = mx + 7$, where m is a constant.

Given that l_1 and l_2 are perpendicular,

- (a) find the value of m .

(2)

The lines l_1 and l_2 meet at the point P .

- (b) Find the x coordinate of P .

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. Find, using algebra, all real solutions to the equation

(i) $16a^2 = 2\sqrt{a}$ (4)

(ii) $b^4 + 7b^2 - 18 = 0$ (4)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. (a) Given that k is a constant, find

$$\int \left(\frac{4}{x^3} + kx \right) dx$$

simplifying your answer.

(3)

- (b) Hence find the value of k such that

$$\int_{0.5}^2 \left(\frac{4}{x^3} + kx \right) dx = 8$$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



4. A tree was planted in the ground.
Its height, H metres, was measured t years after planting.

Exactly 3 years after planting, the height of the tree was 2.35 metres.
Exactly 6 years after planting, the height of the tree was 3.28 metres.

Using a linear model,

- (a) find an equation linking H with t . (3)

The height of the tree was approximately 140 cm when it was planted.

- (b) Explain whether or not this fact supports the use of the linear model in part (a). (2)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



5. A curve has equation

$$y = 3x^2 + \frac{24}{x} + 2 \quad x > 0$$

(a) Find, in simplest form, $\frac{dy}{dx}$ (3)

(b) Hence find the exact range of values of x for which the curve is increasing. (2)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



6.

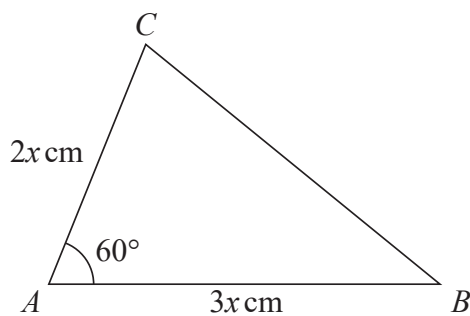
**Figure 1**

Figure 1 shows a sketch of a triangle ABC with $AB = 3x \text{ cm}$, $AC = 2x \text{ cm}$ and angle $CAB = 60^\circ$

Given that the area of triangle ABC is $18\sqrt{3} \text{ cm}^2$

(a) show that $x = 2\sqrt{3}$ (3)

(b) Hence find the exact length of BC , giving your answer as a simplified surd. (3)



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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 6 marks)



7. The curve C has equation

$$y = \frac{k^2}{x} + 1 \quad x \in \mathbb{R}, x \neq 0$$

where k is a constant.

- (a) Sketch C stating the equation of the horizontal asymptote.

(3)

The line l has equation $y = -2x + 5$

- (b) Show that the x coordinate of any point of intersection of l with C is given by a solution of the equation

$$2x^2 - 4x + k^2 = 0$$

(2)

- (c) Hence find the exact values of k for which l is a tangent to C .

(3)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8. (a) Find the first 3 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 + \frac{3x}{4}\right)^6$$

giving each term in its simplest form.

(4)

- (b) Explain how you could use your expansion to estimate the value of 1.925^6
You do not need to perform the calculation.

(1)



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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 5 marks)



9. A company started mining tin in Riverdale on 1st January 2019.

A model to find the total mass of tin that will be mined by the company in Riverdale is given by the equation

$$T = 1200 - 3(n - 20)^2$$

where T tonnes is the total mass of tin mined in the n years after the start of mining.

Using this model,

- (a) calculate the mass of tin that will be mined up to 1st January 2020, (1)
- (b) deduce the maximum total mass of tin that could be mined, (1)
- (c) calculate the mass of tin that will be mined in 2023. (2)
- (d) State, giving reasons, the limitation on the values of n . (2)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



10. A circle C has equation

$$x^2 + y^2 - 4x + 8y - 8 = 0$$

(a) Find

(i) the coordinates of the centre of C ,

(ii) the exact radius of C .

(3)

The straight line with equation $x = k$, where k is a constant, is a tangent to C .

(b) Find the possible values for k .

(2)



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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 5 marks)



11.

$$f(x) = 2x^3 - 13x^2 + 8x + 48$$

- (a) Prove that $(x - 4)$ is a factor of $f(x)$. (2)
- (b) Hence, using algebra, show that the equation $f(x) = 0$ has only two distinct roots. (4)

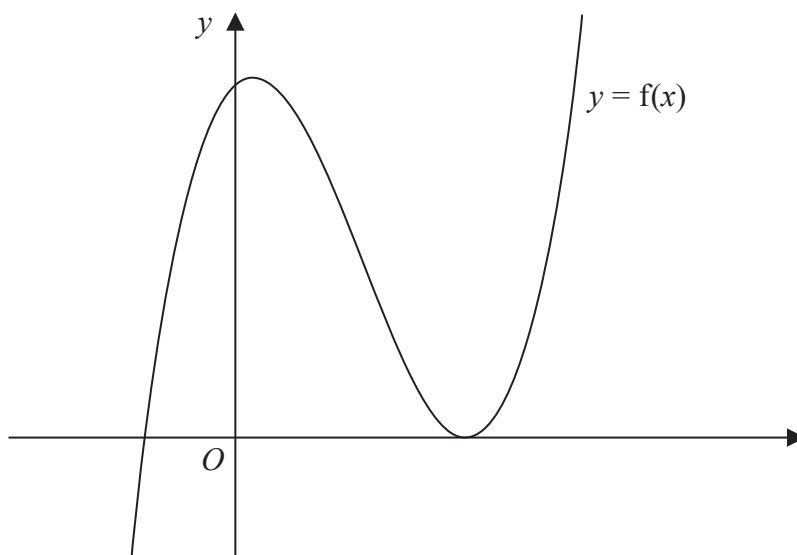


Figure 2

Figure 2 shows a sketch of part of the curve with equation $y = f(x)$.

- (c) Deduce, giving reasons for your answer, the number of real roots of the equation

$$2x^3 - 13x^2 + 8x + 46 = 0 \quad (2)$$

Given that k is a constant and the curve with equation $y = f(x + k)$ passes through the origin,

- (d) find the two possible values of k . (2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 10 marks)



12. (a) Show that

$$\frac{10\sin^2 \theta - 7\cos \theta + 2}{3 + 2\cos \theta} \equiv 4 - 5\cos \theta \quad (4)$$

(b) Hence, or otherwise, solve, for $0 \leq x < 360^\circ$, the equation

$$\frac{10\sin^2 x - 7\cos x + 2}{3 + 2\cos x} = 4 + 3\sin x \quad (3)$$



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 7 marks)



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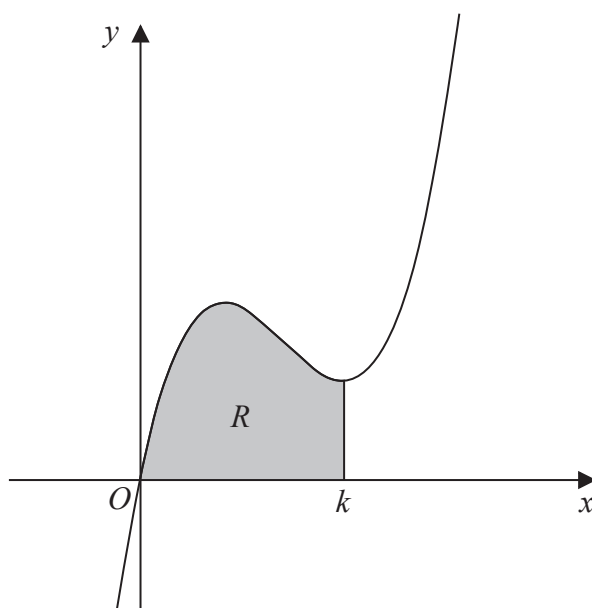
**Figure 3**

Figure 3 shows a sketch of part of the curve with equation

$$y = 2x^3 - 17x^2 + 40x$$

The curve has a minimum turning point at $x = k$.

The region R , shown shaded in Figure 3, is bounded by the curve, the x -axis and the line with equation $x = k$.

Show that the area of R is $\frac{256}{3}$

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(7)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



14. The value of a car, £ V , can be modelled by the equation

$$V = 15\,700e^{-0.25t} + 2300 \quad t \in \mathbb{R}, t \geq 0$$

where the age of the car is t years.

Using the model,

- (a) find the initial value of the car. (1)

Given the model predicts that the value of the car is decreasing at a rate of £500 per year at the instant when $t = T$,

- (b) (i) show that

$$3925e^{-0.25T} = 500$$

- (ii) Hence find the age of the car at this instant, giving your answer in years and months to the nearest month.

(Solutions based entirely on graphical or numerical methods are not acceptable.) (6)

The model predicts that the value of the car approaches, but does not fall below, £ A .

- (c) State the value of A . (1)
- (d) State a limitation of this model. (1)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 9 marks)



15. Given $n \in \mathbb{N}$, prove that $n^3 + 2$ is not divisible by 8

(4)

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Question 15 continued

Lined area for writing the answer to Question 15.

(Total for Question 15 is 4 marks)



16. (i) Two non-zero vectors, **a** and **b**, are such that

$$|\mathbf{a} + \mathbf{b}| = |\mathbf{a}| + |\mathbf{b}|$$

Explain, geometrically, the significance of this statement.

(1)

- (ii) Two different vectors, **m** and **n**, are such that $|\mathbf{m}| = 3$ and $|\mathbf{m} - \mathbf{n}| = 6$
The angle between vector **m** and vector **n** is 30°

Find the angle between vector **m** and vector **m - n**, giving your answer, in degrees, to one decimal place.

(4)



Question 16 continued

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Question 16 continued

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(Total for Question 16 is 5 marks)

(TOTAL FOR PAPER IS 100 MARKS)



Please check the examination details below before entering your candidate information			
Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Wednesday 7 October 2020			
Afternoon (Time: 2 hours)		Paper Reference 8MA0/01	
Mathematics Advanced Subsidiary Paper 1: Pure Mathematics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1/1/1/1/1/



1. A curve has equation

$$y = 2x^3 - 4x + 5$$

Find the equation of the tangent to the curve at the point $P(2, 13)$.

Write your answer in the form $y = mx + c$, where m and c are integers to be found.

Solutions relying on calculator technology are not acceptable.

(5)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



3.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

(i) Solve the equation

$$x\sqrt{2} - \sqrt{18} = x$$

writing the answer as a surd in simplest form.

(3)

(ii) Solve the equation

$$4^{3x-2} = \frac{1}{2\sqrt{2}}$$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 6 marks)



4. In 1997 the average CO₂ emissions of new cars in the UK was 190 g/km.

In 2005 the average CO₂ emissions of new cars in the UK had fallen to 169 g/km.

Given A g/km is the average CO₂ emissions of new cars in the UK n years after 1997 and using a linear model,

(a) form an equation linking A with n .

(3)

In 2016 the average CO₂ emissions of new cars in the UK was 120 g/km.

(b) Comment on the suitability of your model in light of this information.

(3)



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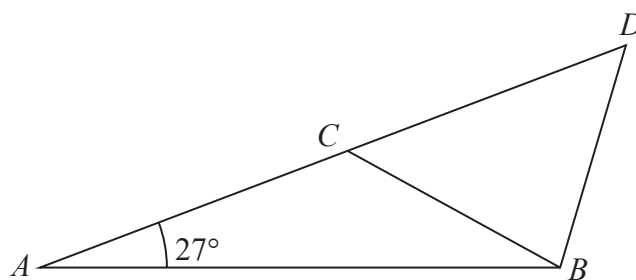
Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



5.



Not to scale

Figure 1

Figure 1 shows the design for a structure used to support a roof.

The structure consists of four steel beams, AB , BD , BC and AD .

Given $AB = 12\text{ m}$, $BC = BD = 7\text{ m}$ and angle $BAC = 27^\circ$

(a) find, to one decimal place, the size of angle ACB .

(3)

The steel beams can only be bought in whole metre lengths.

(b) Find the minimum length of steel that needs to be bought to make the complete structure.

(3)

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 6 marks)



6. (a) Find the first 4 terms, in ascending powers of x , in the binomial expansion of

$$(1 + kx)^{10}$$

where k is a non-zero constant. Write each coefficient as simply as possible.

(3)

Given that in the expansion of $(1 + kx)^{10}$ the coefficient x^3 is 3 times the coefficient of x ,

- (b) find the possible values of k .

(3)



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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 6 marks)



7. Given that k is a positive constant and $\int_1^k \left(\frac{5}{2\sqrt{x}} + 3 \right) dx = 4$

(a) show that $3k + 5\sqrt{k} - 12 = 0$

(4)

(b) Hence, using algebra, find any values of k such that

$$\int_1^k \left(\frac{5}{2\sqrt{x}} + 3 \right) dx = 4$$

(4)



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Question 7 continued

Lined area for writing answers.

(Total for Question 7 is 8 marks)



8. The temperature, $\theta^\circ\text{C}$, of a cup of tea t minutes after it was placed on a table in a room, is modelled by the equation

$$\theta = 18 + 65e^{-\frac{t}{8}} \quad t \geq 0$$

Find, according to the model,

- (a) the temperature of the cup of tea when it was placed on the table, (1)
- (b) the value of t , to one decimal place, when the temperature of the cup of tea was 35°C . (3)
- (c) Explain why, according to this model, the temperature of the cup of tea could not fall to 15°C . (1)

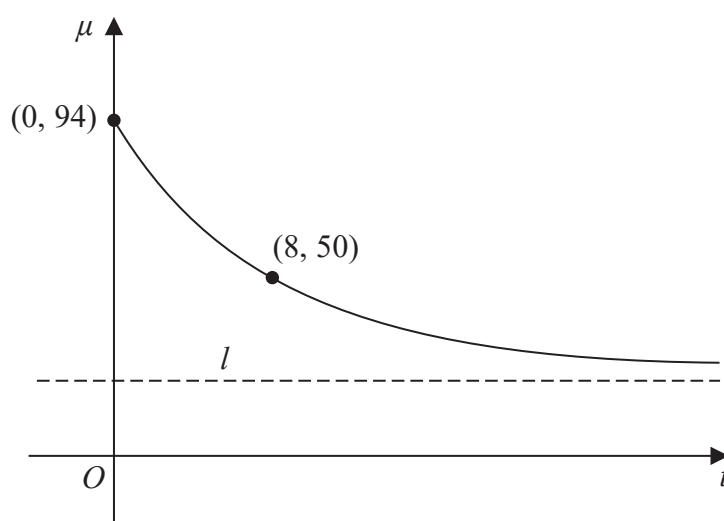


Figure 2

The temperature, $\mu^\circ\text{C}$, of a second cup of tea t minutes after it was placed on a table in a different room, is modelled by the equation

$$\mu = A + Be^{-\frac{t}{8}} \quad t \geq 0$$

where A and B are constants.

Figure 2 shows a sketch of μ against t with two data points that lie on the curve.

The line l , also shown on Figure 2, is the asymptote to the curve.

Using the equation of this model and the information given in Figure 2

- (d) find an equation for the asymptote l . (4)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing answers.

(Total for Question 8 is 9 marks)



9.

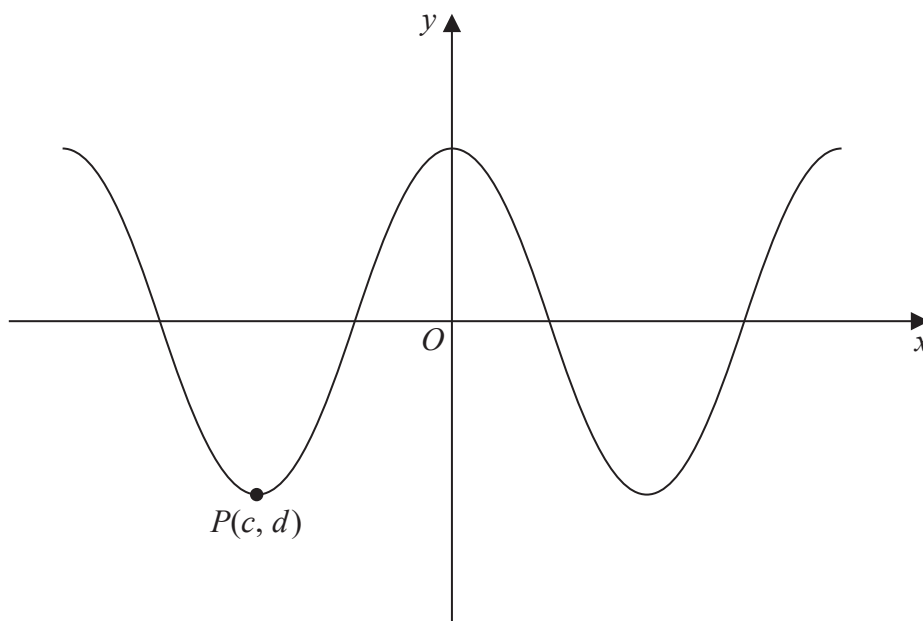


Figure 3

Figure 3 shows part of the curve with equation $y = 3 \cos x^\circ$.

The point $P(c, d)$ is a minimum point on the curve with c being the smallest negative value of x at which a minimum occurs.

(a) State the value of c and the value of d . (1)

(b) State the coordinates of the point to which P is mapped by the transformation which transforms the curve with equation $y = 3 \cos x^\circ$ to the curve with equation

(i) $y = 3 \cos \left(\frac{x^\circ}{4} \right)$

(ii) $y = 3 \cos(x - 36)^\circ$ (2)

(c) Solve, for $450^\circ \leq \theta < 720^\circ$,

$$3 \cos \theta = 8 \tan \theta$$

giving your solution to one decimal place.

In part (c) you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable. (5)



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 8 marks)



10.

$$g(x) = 2x^3 + x^2 - 41x - 70$$

- (a) Use the factor theorem to show that $g(x)$ is divisible by $(x - 5)$. **(2)**
- (b) Hence, showing all your working, write $g(x)$ as a product of three linear factors. **(4)**

The finite region R is bounded by the curve with equation $y = g(x)$ and the x -axis, and lies below the x -axis.

- (c) Find, using algebraic integration, the exact value of the area of R . (4)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 10 marks)



11. (i) A circle C_1 has equation

$$x^2 + y^2 + 18x - 2y + 30 = 0$$

The line l is the tangent to C_1 at the point $P(-5, 7)$.

Find an equation of l in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(5)

- (ii) A different circle C_2 has equation

$$x^2 + y^2 - 8x + 12y + k = 0$$

where k is a constant.

Given that C_2 lies entirely in the 4th quadrant, find the range of possible values for k .

(4)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 9 marks)



12. An advertising agency is monitoring the number of views of an online advert.

The equation

$$\log_{10} V = 0.072t + 2.379 \quad 1 \leq t \leq 30, t \in \mathbb{N}$$

is used to model the total number of views of the advert, V , in the first t days after the advert went live.

- (a) Show that $V = ab^t$ where a and b are constants to be found.

Give the value of a to the nearest whole number and give the value of b to 3 significant figures.

(4)

- (b) Interpret, with reference to the model, the value of ab .

(1)

Using this model, calculate

- (c) the total number of views of the advert in the first 20 days after the advert went live.
Give your answer to 2 significant figures.

(2)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 7 marks)



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13. (a) Prove that for all positive values of a and b

$$\frac{4a}{b} + \frac{b}{a} \geq 4 \quad (4)$$

(b) Prove, by counter example, that this is not true for all values of a and b . (1)



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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 5 marks)



14. A curve has equation $y = g(x)$.

Given that

- $g(x)$ is a cubic expression in which the coefficient of x^3 is equal to the coefficient of x
- the curve with equation $y = g(x)$ passes through the origin
- the curve with equation $y = g(x)$ has a stationary point at $(2, 9)$

(a) find $g(x)$,

(7)

(b) prove that the stationary point at $(2, 9)$ is a maximum.

(2)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 9 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 2 hours	Paper reference	8MA0/01	
Mathematics Advanced Subsidiary PAPER 1: Pure Mathematics			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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1. In this question you should show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Using algebra, solve the inequality

$$x^2 - x > 20$$

writing your answer in set notation.

(3)



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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 3 marks)



2.

In this question you should show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

Given

$$\frac{9^{x-1}}{3^{y+2}} = 81$$

express y in terms of x , writing your answer in simplest form.

(3)



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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 3 marks)



3. Find

$$\int \frac{3x^4 - 4}{2x^3} dx$$

writing your answer in simplest form.

(4)

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 4 marks)



4. [In this question the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are due east and due north respectively.]

A stone slides horizontally across ice.

Initially the stone is at the point $A(-24\mathbf{i} - 10\mathbf{j})$ m relative to a fixed point O .

After 4 seconds the stone is at the point $B(12\mathbf{i} + 5\mathbf{j})$ m relative to the fixed point O .

The motion of the stone is modelled as that of a particle moving in a straight line at constant speed.

Using the model,

- (a) prove that the stone passes through O ,

(2)

- (b) calculate the speed of the stone.

(3)



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Question 4 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 4 is 5 marks)



5.

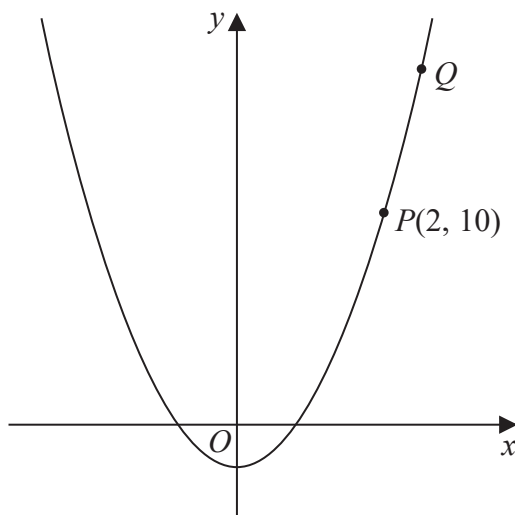


Figure 1

Figure 1 shows part of the curve with equation $y = 3x^2 - 2$

The point $P(2, 10)$ lies on the curve.

- (a) Find the gradient of the tangent to the curve at P . (2)

The point Q with x coordinate $2 + h$ also lies on the curve.

- (b) Find the gradient of the line PQ , giving your answer in terms of h in simplest form. **(3)**
- (c) Explain briefly the relationship between part (b) and the answer to part (a). **(1)**



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6. In this question you should show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Using algebra, find all solutions of the equation

$$3x^3 - 17x^2 - 6x = 0 \quad (3)$$

(b) Hence find all real solutions of

$$3(y-2)^6 - 17(y-2)^4 - 6(y-2)^2 = 0 \quad (3)$$



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7. A parallelogram $PQRS$ has area 50 cm^2

Given

- PQ has length 14 cm
- QR has length 7 cm
- angle SPQ is obtuse

find

(a) the size of angle SPQ , in degrees, to 2 decimal places,

(3)

(b) the length of the diagonal SQ , in cm, to one decimal place.

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 5 marks)



8. $g(x) = (2 + ax)^8$ where a is a constant

Given that one of the terms in the binomial expansion of $g(x)$ is $3402x^5$

(a) find the value of a .

(4)

Using this value of a ,

(b) find the constant term in the expansion of

$$\left(1 + \frac{1}{x^4}\right)(2 + ax)^8$$

(3)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 7 marks)



9. Find the value of the constant k , $0 < k < 9$, such that

$$\int_k^9 \frac{6}{\sqrt{x}} dx = 20$$

(4)

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 4 marks)



10. A student is investigating the following statement about natural numbers.

“ $n^3 - n$ is a multiple of 4”

(a) Prove, using algebra, that the statement is true for all odd numbers.

(4)

(b) Use a counterexample to show that the statement is not always true.

(1)



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Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 5 marks)



- 11.** The owners of a nature reserve decided to increase the area of the reserve covered by trees.

Tree planting started on 1st January 2005.

The area of the nature reserve covered by trees, $A \text{ km}^2$, is modelled by the equation

$$A = 80 - 45e^{ct}$$

where c is a constant and t is the number of years after 1st January 2005.

Using the model,

- (a) find the area of the nature reserve that was covered by trees just before tree planting started. (1)

On 1st January 2019 an area of 60km^2 of the nature reserve was covered by trees.

- (b) Use this information to find a complete equation for the model, giving your value of c to 3 significant figures. (4)

On 1st January 2020, the owners of the nature reserve announced a long-term plan to have 100 km² of the nature reserve covered by trees.

- (c) State a reason why the model is not appropriate for this plan. (1)



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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 6 marks)



12.

In this question you should show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for
- $0 < \theta \leq 450^\circ$
- , the equation

$$5 \cos^2 \theta = 6 \sin \theta$$

giving your answers to one decimal place.

(5)

- (ii) (a) A student's attempt to solve the question

“Solve, for $-90^\circ < x < 90^\circ$, the equation $3 \tan x - 5 \sin x = 0$ ”

is set out below.

$$\begin{aligned} 3 \tan x - 5 \sin x &= 0 \\ 3 \frac{\sin x}{\cos x} - 5 \sin x &= 0 \\ 3 \sin x - 5 \sin x \cos x &= 0 \\ 3 - 5 \cos x &= 0 \\ \cos x &= \frac{3}{5} \\ x &= 53.1^\circ \end{aligned}$$

Identify two errors or omissions made by this student, giving a brief explanation of each.

(2)

The first four positive solutions, in order of size, of the equation

$$\cos(5\alpha + 40^\circ) = \frac{3}{5}$$

are $\alpha_1, \alpha_2, \alpha_3$ and α_4

- (b) Find, to the nearest degree, the value of
- α_4

(2)



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Question 12 continued

Handwriting practice area with horizontal lines.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 9 marks)



13.

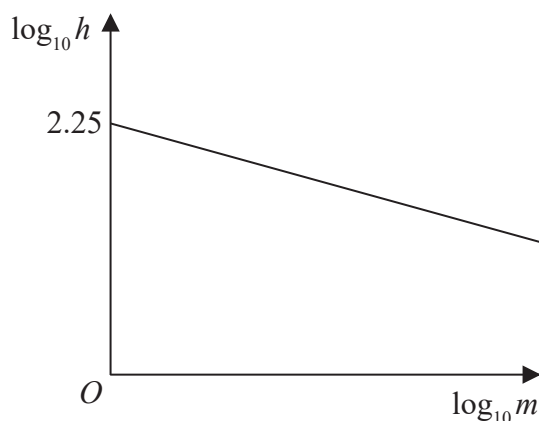


Figure 2

The resting heart rate, h , of a mammal, measured in beats per minute, is modelled by the equation

$$h = pm^q$$

where p and q are constants and m is the mass of the mammal measured in kg.

Figure 2 illustrates the linear relationship between $\log_{10} h$ and $\log_{10} m$

The line meets the vertical $\log_{10} h$ axis at 2.25 and has a gradient of -0.235

- (a) Find, to 3 significant figures, the value of p and the value of q . (3)

A particular mammal has a mass of 5 kg and a resting heart rate of 119 beats per minute.

- (b) Comment on the suitability of the model for this mammal. (3)

- (c) With reference to the model, interpret the value of the constant p . (1)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



14. A curve C has equation $y = f(x)$ where

$$f(x) = -3x^2 + 12x + 8$$

(a) Write $f(x)$ in the form

$$a(x + b)^2 + c$$

where a , b and c are constants to be found.

(3)

The curve C has a maximum turning point at M .

(b) Find the coordinates of M .

(2)

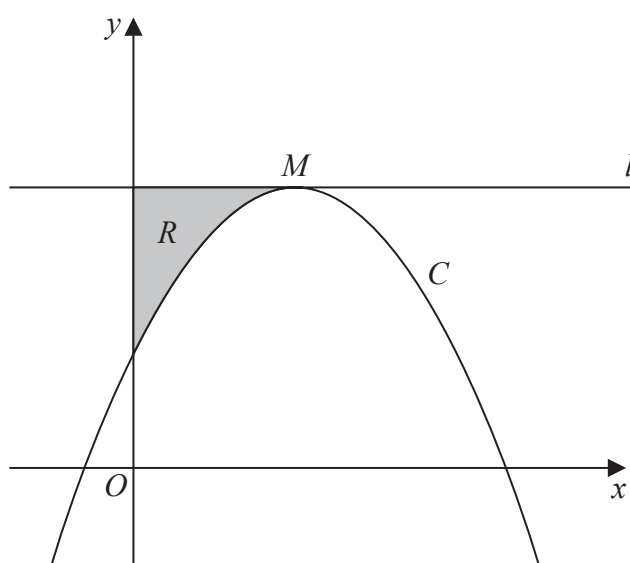


Figure 3

Figure 3 shows a sketch of the curve C .

The line l passes through M and is parallel to the x -axis.

The region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

(c) Using algebraic integration, find the area of R .

(5)



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Question 14 continued

Handwriting practice area with horizontal lines.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 10 marks)



15.

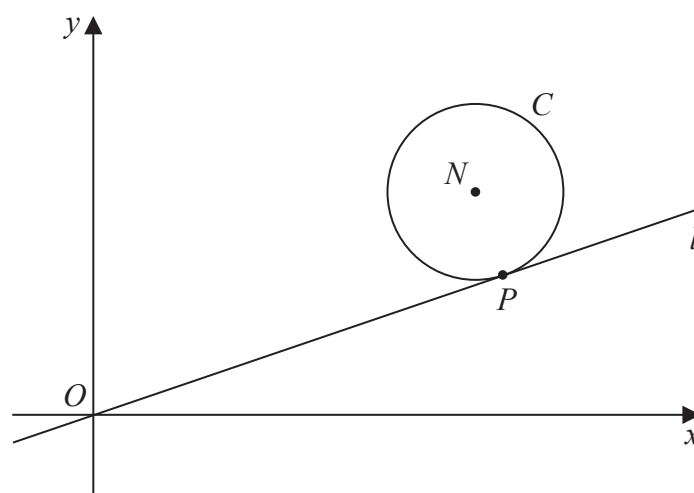


Figure 4

Figure 4 shows a sketch of a circle C with centre $N(7, 4)$

The line l with equation $y = \frac{1}{3}x$ is a tangent to C at the point P .

Find

(a) the equation of line PN in the form $y = mx + c$, where m and c are constants, (2)

(b) an equation for C . (4)

The line with equation $y = \frac{1}{3}x + k$, where k is a non-zero constant, is also a tangent to C .

(c) Find the value of k . (3)



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Question 15 continued

Handwriting practice area with 30 horizontal lines.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.

(Total for Question 15 is 9 marks)



16. The curve C has equation $y = f(x)$ where

$$f(x) = ax^3 + 15x^2 - 39x + b$$

and a and b are constants.

Given

- the point $(2, 10)$ lies on C
- the gradient of the curve at $(2, 10)$ is -3

(a) (i) show that the value of a is -2

(ii) find the value of b .

(4)

(b) Hence show that C has no stationary points.

(3)

(c) Write $f(x)$ in the form $(x - 4)Q(x)$ where $Q(x)$ is a quadratic expression to be found.

(2)

(d) Hence deduce the coordinates of the points of intersection of the curve with equation

$$y = f(0.2x)$$

and the coordinate axes.

(2)



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Question 16 continued

Handwriting practice area with horizontal lines.



Question 16 continued

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(Total for Question 16 is 11 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Time 2 hours

Paper reference **8MA0/01**

Mathematics

Advanced Subsidiary

PAPER 1: Pure Mathematics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Q:1/1/1/1/




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1. Find

$$\int \left(8x^3 - \frac{3}{2\sqrt{x}} + 5 \right) dx$$

giving your answer in simplest form.

(4)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2.

$$f(x) = 2x^3 + 5x^2 + 2x + 15$$

(a) Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$. (2)

(b) Find the constants a , b and c such that

$$f(x) = (x + 3)(ax^2 + bx + c) \quad (2)$$

(c) Hence show that $f(x) = 0$ has only one real root. (2)

(d) Write down the real root of the equation $f(x - 5) = 0$ (1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3. The triangle PQR is such that $\vec{PQ} = 3\mathbf{i} + 5\mathbf{j}$ and $\vec{PR} = 13\mathbf{i} - 15\mathbf{j}$

(a) Find $\vec{QR}$ (2)

(b) Hence find $|\vec{QR}|$ giving your answer as a simplified surd. (2)

The point S lies on the line segment QR so that $QS:SR = 3:2$

(c) Find $\vec{PS}$ (2)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 6 marks)



4.

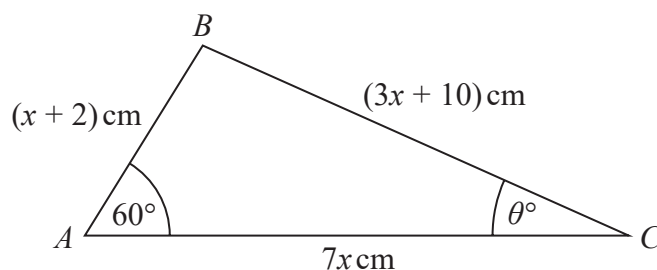
**Figure 1**

Figure 1 shows a sketch of triangle ABC with $AB = (x + 2) \text{ cm}$, $BC = (3x + 10) \text{ cm}$, $AC = 7x \text{ cm}$, angle $BAC = 60^\circ$ and angle $ACB = \theta^\circ$

(a) (i) Show that $17x^2 - 35x - 48 = 0$ (3)

(ii) Hence find the value of x . (1)

(b) Hence find the value of θ giving your answer to one decimal place. (2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



- 5.** The mass, A kg, of algae in a small pond, is modelled by the equation

$$A = pq^t$$

where p and q are constants and t is the number of weeks after the mass of algae was first recorded.

Data recorded indicates that there is a linear relationship between t and $\log_{10} A$ given by the equation

$$\log_{10} A = 0.03t + 0.5$$

- (a) Use this relationship to find a complete equation for the model in the form

$$A = pq^t$$

giving the value of p and the value of q each to 4 significant figures.

(4)

- (b) With reference to the model, interpret

- (i) the value of the constant p ,
- (ii) the value of the constant q .

(2)

- (c) Find, according to the model,

- (i) the mass of algae in the pond when $t = 8$, giving your answer to the nearest 0.5 kg,
- (ii) the number of weeks it takes for the mass of algae in the pond to reach 4 kg.

(3)

- (d) State one reason why this may not be a realistic model in the long term.

(1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



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6. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(3 - \frac{2x}{9}\right)^8$$

giving each term in simplest form.

(4)

$$f(x) = \left(\frac{x-1}{2x}\right)\left(3 - \frac{2x}{9}\right)^8$$

- (b) Find the coefficient of x^2 in the series expansion of $f(x)$, giving your answer as a simplified fraction.

(2)

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7. (a) Factorise completely $9x - x^3$ (2)

The curve C has equation

$$y = 9x - x^3$$

- (b) Sketch C showing the coordinates of the points at which the curve cuts the x -axis. (2)

The line l has equation $y = k$ where k is a constant.

Given that C and l intersect at 3 distinct points,

- (c) find the range of values for k , writing your answer in set notation.

Solutions relying on calculator technology are not acceptable. (3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 7 marks)



8.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

The air pressure, P kg/cm², inside a car tyre, t minutes from the instant when the tyre developed a puncture is given by the equation

$$P = k + 1.4e^{-0.5t} \quad t \in \mathbb{R} \quad t \geq 0$$

where k is a constant.

Given that the initial air pressure inside the tyre was 2.2 kg/cm²

(a) state the value of k .

(1)

From the instant when the tyre developed the puncture,

(b) find the time taken for the air pressure to fall to 1 kg/cm²
Give your answer in minutes to one decimal place.

(3)

(c) Find the rate at which the air pressure in the tyre is decreasing exactly 2 minutes from the instant when the tyre developed the puncture.
Give your answer in kg/cm² per minute to 3 significant figures.

(2)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 6 marks)



9. (a) Given that $p = \log_3 x$, where $x > 0$, find in simplest form in terms of p ,

(i) $\log_3 \left(\frac{x}{9} \right)$

(ii) $\log_3 (\sqrt{x})$

(2)

(b) Hence, or otherwise, solve

$$2 \log_3 \left(\frac{x}{9} \right) + 3 \log_3 (\sqrt{x}) = -11$$

giving your answer as a simplified fraction.

Solutions relying on calculator technology are not acceptable.

(4)



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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



10.

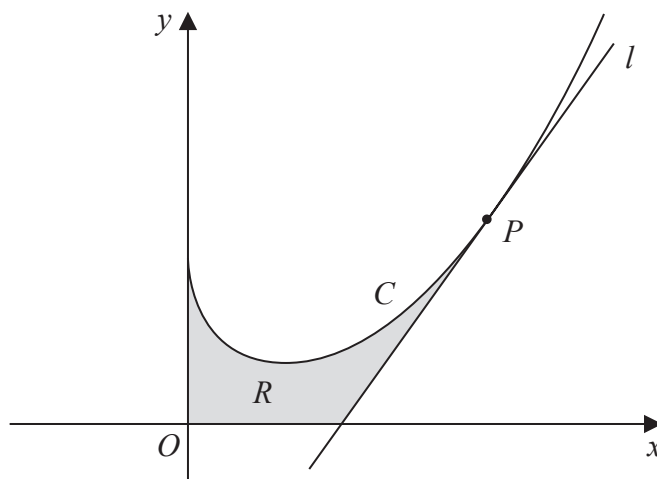


Figure 2

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{1}{3}x^2 - 2\sqrt{x} + 3 \quad x \geq 0$$

The point P lies on C and has x coordinate 4

The line l is the tangent to C at P .

(a) Show that l has equation

$$13x - 6y - 26 = 0 \quad (5)$$

The region R , shown shaded in Figure 2, is bounded by the y -axis, the curve C , the line l and the x -axis.

(b) Find the exact area of R . (5)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 10 marks)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 8 marks)



12. A company makes drinks containers out of metal.

The containers are modelled as closed cylinders with base radius r cm and height h cm and the capacity of each container is 355 cm^3

The metal used

- for the circular base and the curved side costs 0.04 pence/cm²
- for the circular top costs 0.09 pence/cm²

Both metals used are of negligible thickness.

- (a) Show that the total cost, C pence, of the metal for one container is given by

$$C = 0.13\pi r^2 + \frac{28.4}{r} \quad (4)$$

- (b) Use calculus to find the value of r for which C is a minimum, giving your answer to 3 significant figures.

- (c) Using $\frac{d^2C}{dr^2}$ prove that the cost is minimised for the value of r found in part (b).

- (d) Hence find the minimum value of C , giving your answer to the nearest integer.



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 12 marks)



13.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that

$$\frac{1}{\cos \theta} + \tan \theta \equiv \frac{\cos \theta}{1 - \sin \theta} \quad \theta \neq (2n + 1)90^\circ \quad n \in \mathbb{Z} \quad (3)$$

Given that $\cos 2x \neq 0$ (b) solve for $0 < x < 90^\circ$

$$\frac{1}{\cos 2x} + \tan 2x = 3 \cos 2x$$

giving your answers to one decimal place.

(5)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 8 marks)



14. (i) A student states

“if x^2 is greater than 9 then x must be greater than 3”

Determine whether or not this statement is true, giving a reason for your answer.

(1)

(ii) Prove that for all positive integers n ,

$$n^3 + 3n^2 + 2n$$

is divisible by 6

(3)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Thursday 18 May 2023

Afternoon (Time: 2 hours)

Paper reference **8MA0/01**

Mathematics

Advanced Subsidiary

PAPER 1: Pure Mathematics

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear.
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Information

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- There are 17 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
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Advice

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- Try to answer every question.
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Turn over ►

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1. A curve has equation

$$y = \frac{2}{3}x^3 - \frac{7}{2}x^2 - 4x + 5$$

(a) Find $\frac{dy}{dx}$ writing your answer in simplest form.

(2)

(b) Hence find the range of values of x for which y is decreasing.

(4)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Using the substitution $u = \sqrt{x}$ or otherwise, solve

$$6x + 7\sqrt{x} - 20 = 0$$

(4)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3.

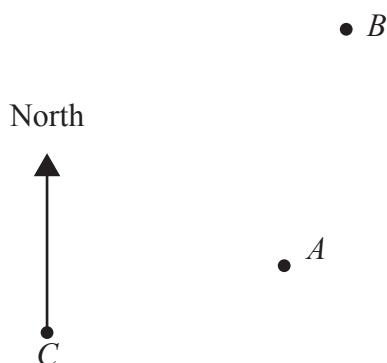


Figure 1

Figure 1 is a sketch showing the position of three phone masts, A , B and C .

The masts are identical and their bases are assumed to lie in the same horizontal plane.

From mast C

- mast A is 8.2 km away on a bearing of 072°
- mast B is 15.6 km away on a bearing of 039°

(a) Find the distance between masts A and B , giving your answer in km to one decimal place.

(3)

An engineer needs to travel from mast A to mast B .

(b) Give a reason why the answer to part (a) is unlikely to be an accurate value for the distance the engineer travels.

(1)

[illegible]

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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 4 marks)



4. (a) Sketch the curve with equation

$$y = \frac{k}{x} \quad x \neq 0$$

where k is a positive constant.

(2)

- (b) Hence or otherwise, solve

$$\frac{16}{x} \leq 2$$

(3)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



5.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

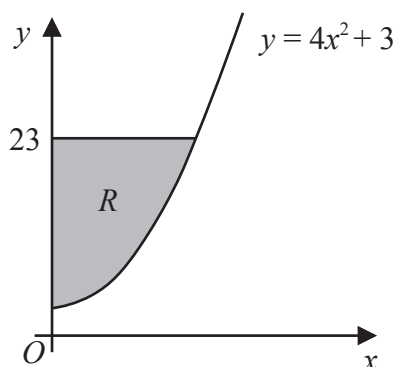


Figure 2

The finite region R , shown shaded in Figure 2, is bounded by the curve with equation $y = 4x^2 + 3$, the y -axis and the line with equation $y = 23$

Show that the exact area of R is $k\sqrt{5}$ where k is a rational constant to be found.

(5)

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 5 marks)



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6. The circle C has equation

$$x^2 + y^2 - 6x + 10y + k = 0$$

where k is a constant.

- (a) Find the coordinates of the centre of C .

(2)

Given that C does not cut or touch the x -axis,

- (b) find the range of possible values for k .

(3)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 5 marks)



7. The distance a particular car can travel in a journey starting with a full tank of fuel was investigated.
- From a full tank of fuel, 40 litres remained in the car's fuel tank after the car had travelled 80 km
 - From a full tank of fuel, 25 litres remained in the car's fuel tank after the car had travelled 200 km

Using a **linear model**, with V litres being the volume of fuel remaining in the car's fuel tank and d km being the distance the car had travelled,

- (a) find an equation linking V with d . (4)

Given that, on a particular journey

- the fuel tank of the car was initially full
- the car continued until it ran out of fuel

find, according to the model,

- (b) (i) the initial volume of fuel that was in the fuel tank of the car,
(ii) the distance that the car travelled on this journey.
- (3)**

In fact the car travelled 320 km on this journey.

- (c) Evaluate the model in light of this information. (1)

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8.

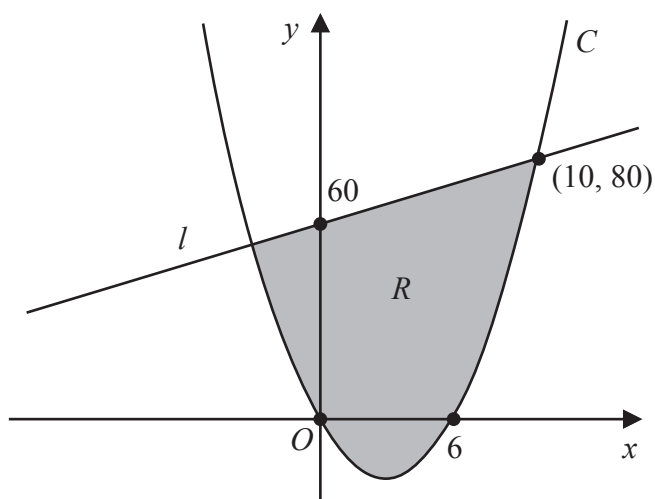


Figure 3

Figure 3 shows a sketch of a curve C and a straight line l .

Given that

- C has equation $y = f(x)$ where $f(x)$ is a quadratic expression in x
- C cuts the x -axis at 0 and 6
- l cuts the y -axis at 60 and intersects C at the point $(10, 80)$

use inequalities to define the region R shown shaded in Figure 3.

(5)



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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 5 marks)



9. Using the laws of logarithms, solve the equation

$$2\log_5(3x - 2) - \log_5 x = 2$$

(5)



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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 5 marks)



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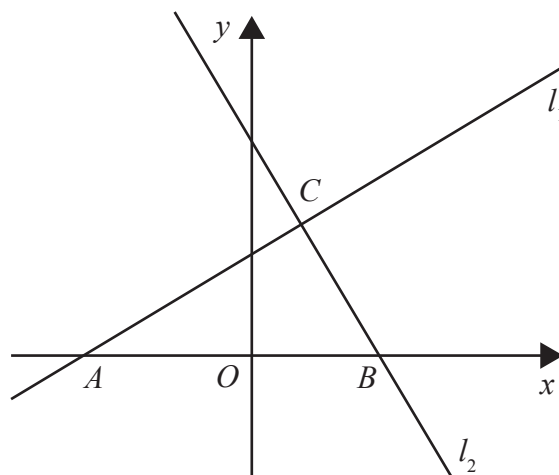


Figure 4

The line l_1 has equation $y = \frac{3}{5}x + 6$

The line l_2 is perpendicular to l_1 and passes through the point $B(8,0)$, as shown in the sketch in Figure 4.

(a) Show that an equation for line l_2 is

$$5x + 3y = 40 \tag{3}$$

Given that

- lines l_1 and l_2 intersect at the point C
- line l_1 crosses the x -axis at the point A

(b) find the exact area of triangle ABC , giving your answer as a fully simplified fraction in the form $\frac{p}{q}$



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 6 marks)



12.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

- (a) Show that the equation

$$4 \tan x = 5 \cos x$$

can be written as

$$5\sin^2 x + 4\sin x - 5 = 0 \quad (3)$$

- (b) Hence solve, for $0 < x \leq 360^\circ$

$$4 \tan x = 5 \cos x$$

giving your answers to one decimal place. (4)

- (c) Hence find the **number of solutions** of the equation

$$4 \tan 3x = 5 \cos 3x$$

in the interval $0 < x \leq 1800^\circ$, explaining briefly the reason for your answer. (2)

[illegible]

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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 9 marks)



13. Relative to a fixed origin O

- point A has position vector $10\mathbf{i} - 3\mathbf{j}$
- point B has position vector $-8\mathbf{i} + 9\mathbf{j}$
- point C has position vector $-2\mathbf{i} + p\mathbf{j}$ where p is a constant

(a) Find $\overrightarrow{AB}$

(b) Find $|\overrightarrow{AB}|$ giving your answer as a fully simplified surd. (2)

Given that points A , B and C lie on a straight line,

(c) (i) find the value of p ,
(ii) state the ratio of the area of triangle AOC to the area of triangle AOB .

(3)

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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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DO NOT WRITE IN THIS AREA

Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 5 marks)



15.

In this question you must show detailed reasoning.**Solutions relying on calculator technology are not acceptable.**

The curve C_1 has equation $y = 8 - 10x + 6x^2 - x^3$

The curve C_2 has equation $y = x^2 - 12x + 14$

(a) Verify that when $x = 1$ the curves C_1 and C_2 intersect.

(2)

The curves also intersect when $x = k$.

Given that $k < 0$

(b) use algebra to find the exact value of k .

(5)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Handwriting practice area with horizontal lines.

(Total for Question 15 is 7 marks)



16. A curve has equation $y = f(x)$, $x \geq 0$

Given that

- $f'(x) = 4x + a\sqrt{x} + b$, where a and b are constants
- the curve has a stationary point at $(4, 3)$
- the curve meets the y -axis at -5

find $f(x)$, giving your answer in simplest form.

(6)



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Question 16 continued

Lined area for writing the answer to Question 16.

(Total for Question 16 is 6 marks)



17. In this question p and q are positive integers with $q > p$

Statement 1: $q^3 - p^3$ is never a multiple of 5

(a) Show, by means of a counter example, that Statement 1 is **not** true.

(1)

Statement 2: When p and q are consecutive **even** integers $q^3 - p^3$ is a multiple of 8

(b) Prove, using algebra, that Statement 2 is true.

(4)



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Question 17 continued

Lined area for writing the answer to Question 17.



Question 17 continued

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(Total for Question 17 is 5 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Thursday 16 May 2024

Afternoon (Time: 2 hours)

Paper reference **8MA0/01**

Mathematics

Advanced Subsidiary

PAPER 1: Pure Mathematics

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Turn over ►

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1. Find

$$\int \frac{2\sqrt{x} - 3}{x^2} dx$$

giving your answer in simplest form.

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 4 marks)



2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

$$f(x) = 2x^3 - 3ax^2 + bx + 8a$$

where a and b are constants.

Given that $(x - 4)$ is a factor of $f(x)$,

(a) use the factor theorem to show that

$$10a = 32 + b \quad (2)$$

Given also that $(x - 2)$ is a factor of $f(x)$,

(b) express $f(x)$ in the form

$$f(x) = (2x + k)(x - 4)(x - 2) \quad (4)$$

where k is a constant to be found.

(c) Hence,

(i) state the number of real roots of the equation $f(x) = 0$

(ii) write down the largest root of the equation $f\left(\frac{1}{3}x\right) = 0 \quad (2)$



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 8 marks)

3. Relative to a fixed origin O ,

- point P has position vector $9\mathbf{i} - 8\mathbf{j}$
- point Q has position vector $3\mathbf{i} - 5\mathbf{j}$

(a) Find $\overrightarrow{PQ}$ (2)

Given that R is the point such that $\overrightarrow{QR} = 9\mathbf{i} + 18\mathbf{j}$

(b) show that angle $PQR = 90^\circ$ (2)

Given also that S is the point such that $\overrightarrow{PS} = 3\overrightarrow{QR}$

(c) find the exact area of $PQRS$ (4)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 8 marks)

4.

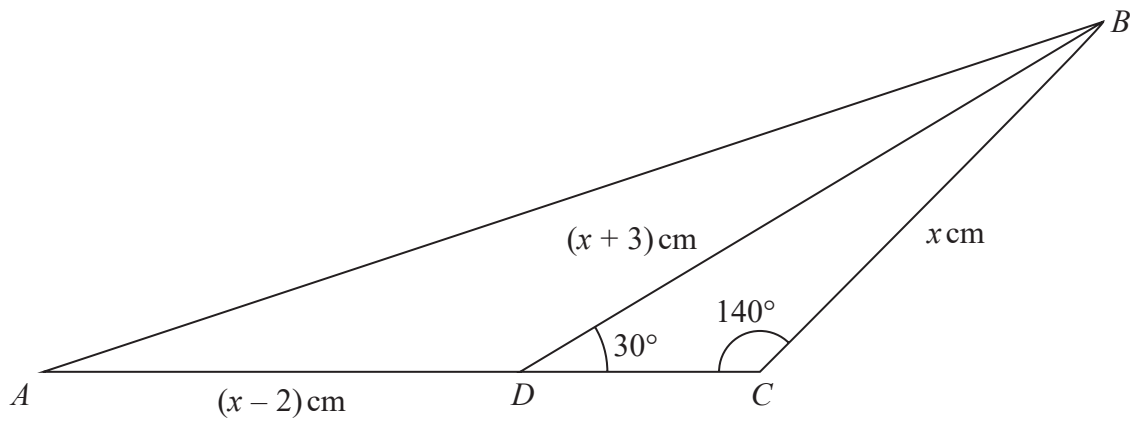


Figure 1

Figure 1 shows a sketch of triangle ABD and triangle BCD

Given that

- ADC is a straight line
- $BD = (x + 3)$ cm
- $BC = x$ cm
- $\text{angle } BDC = 30^\circ$
- $\text{angle } BCD = 140^\circ$

(a) show that $x = 10.5$ correct to 3 significant figures.

(3)

Given also that $AD = (x - 2)$ cm

(b) find the length of AB , giving your answer to 3 significant figures.

(2)



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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 5 marks)



5. The curve C_1 has equation

$$y = \frac{6}{x} + 3$$

(a) (i) Sketch C_1 stating the coordinates of any points where the curve cuts the coordinate axes.

(ii) State the equations of any asymptotes to the curve C_1

(3)

The curve C_2 has equation

$$y = 3x^2 - 4x - 10$$

(b) Show that C_1 and C_2 intersect when

$$3x^3 - 4x^2 - 13x - 6 = 0$$

(2)

Given that the x coordinate of one of the points of intersection is $-\frac{2}{3}$

(c) use algebra to find the x coordinates of the other points of intersection between C_1 and C_2

(Solutions relying on calculator technology are not acceptable.)

(4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 9 marks)



6. The binomial expansion of

$$(1 + ax)^{12}$$

up to and including the term in x^2 is

$$1 - \frac{15}{2}x + kx^2$$

where a and k are constants.

(a) Show that $a = -\frac{5}{8}$

(2)

(b) Hence find the value of k

(2)

Using the expansion and making your method clear,

(c) find an estimate for the value of $\left(\frac{17}{16}\right)^{12}$, giving your answer to 4 decimal places.

(2)

[illegible]

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Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 6 marks)



DO NOT WRITE IN THIS AREA

Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 5 marks)



8.

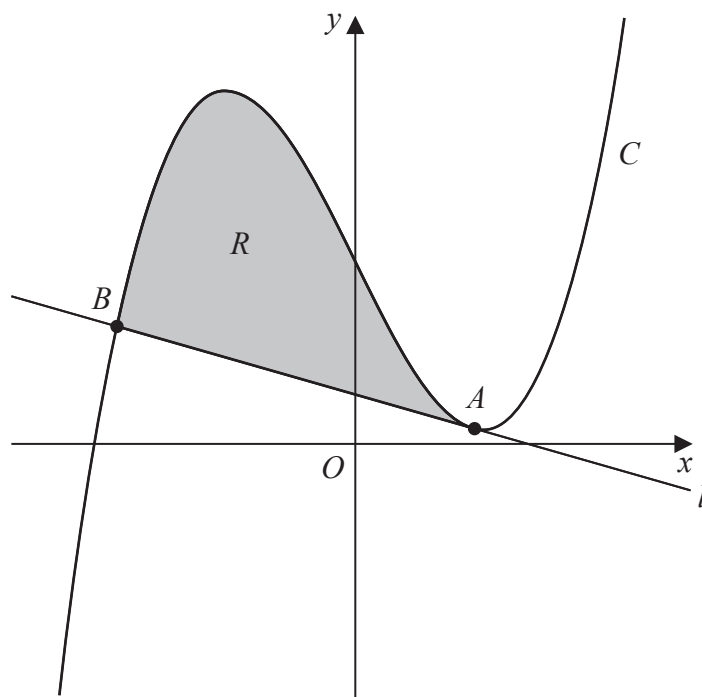


Figure 3

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Figure 3 shows a sketch of the curve C with equation

$$y = x^3 - 14x + 23$$

The line l is the tangent to C at the point A , also shown in Figure 3.

Given that l has equation $y = -2x + 7$

- (a) show, using calculus, that the x coordinate of A is 2 (3)

The line l cuts C again at the point B .

- (b) Verify that the x coordinate of B is -4 (2)

The finite region, R , shown shaded in Figure 3, is bounded by C and l .

Using algebraic integration,

- (c) show that the area of R is 108 (5)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 10 marks)



9.

$$p = \log_a 16$$

$$q = \log_a 25$$

where a is a constant.

Find in terms of p and/or q ,

(a) $\log_a 256$ (1)

(b) $\log_a 100$ (2)

(c) $\log_a 80 \times \log_a 3.2$ (2)



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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 5 marks)



10.

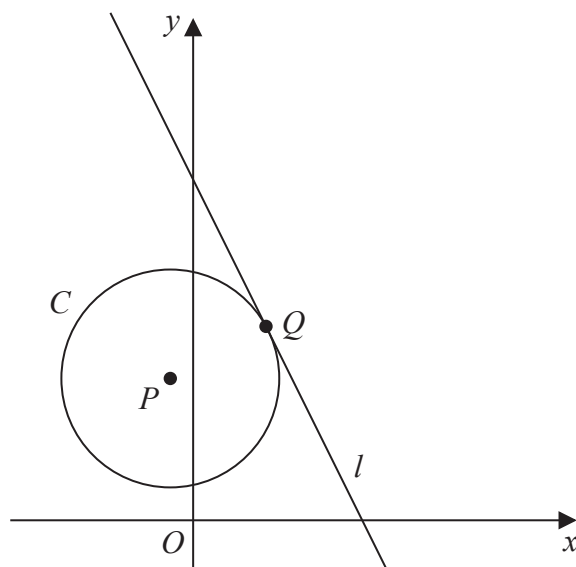


Figure 4

Figure 4 shows a sketch of the circle C

- the point $P(-1, k+8)$ is the centre of C
- the point $Q(3, k^2-2k)$ lies on C
- k is a positive constant
- the line l is the tangent to C at Q

Given that the gradient of l is -2

(a) show that

$$k^2 - 3k - 10 = 0 \tag{4}$$

(b) Hence find an equation for C (4)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



11. The prices of two precious metals are being monitored.

The price per gram of metal A , £ V_A , is modelled by the equation

$$V_A = 100 + 20e^{0.04t}$$

where t is the number of months after monitoring began.

The price per gram of metal B , £ V_B , is modelled by the equation

$$V_B = pe^{-0.02t}$$

where p is a positive constant and t is the number of months after monitoring began.

Given that $V_B = 2V_A$ when $t = 0$

(a) find the value of p

(2)

When $t = T$, the rate of **increase** in the price per gram of metal A was equal to the rate of **decrease** in the price per gram of metal B

(b) Find the value of T , giving your answer to one decimal place.

(Solutions based entirely on calculator technology are not acceptable.)

(4)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 6 marks)



12.

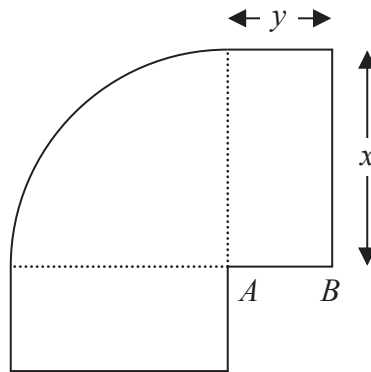


Figure 5

Figure 5 shows the plan view of the design for a swimming pool.

The pool is modelled as a quarter of a circle joined to two equal sized rectangles as shown.

Given that

- the quarter circle has radius x metres
- the rectangles each have length x metres and width y metres
- the total surface area of the swimming pool is 100 m^2

- (a) show that, according to the model, the perimeter P metres of the swimming pool is given by

$$P = 2x + \frac{200}{x} \quad (5)$$

- (b) Use calculus to find the value of x for which P has a stationary value. (4)

- (c) Prove, by further calculus, that this value of x gives a minimum value for P (2)

Access to the pool is by side AB shown in Figure 5.

Given that AB must be at least one metre,

- (d) determine, according to the model, whether the swimming pool with the minimum perimeter would be suitable. (2)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 13 marks)



13.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$\sin \theta(7 \sin \theta - 4 \cos \theta) = 4$$

can be written as

$$3 \tan^2 \theta - 4 \tan \theta - 4 = 0 \quad (4)$$

(b) Hence solve, for $0 < x < 360^\circ$

$$\sin x(7 \sin x - 4 \cos x) = 4$$

giving your answers to one decimal place.

(c) Hence find the smallest solution of the equation

$$\sin 4\alpha(7 \sin 4\alpha - 4 \cos 4\alpha) = 4$$

°, giving your answer to one decimal place.

(1)

in the range $720^\circ < \alpha < 1080^\circ$, giving your answer to one decimal place.



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 9 marks)



14. Prove, using algebra, that

$$n^2 + 5n$$

is even for all $n \in \mathbb{N}$

(4)

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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Thursday 15 May 2025

Afternoon (Time: 2 hours)

Paper reference **8MA0/01**

Mathematics
Advanced Subsidiary
PAPER 1: Pure Mathematics

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1.

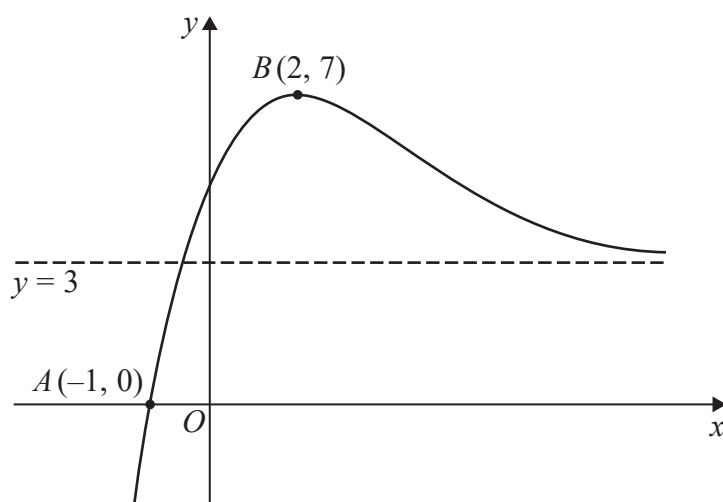
**Figure 1**

Figure 1 shows a curve with equation $y = f(x)$

The curve

- passes through the point $A(-1, 0)$
- has a maximum turning point at $B(2, 7)$
- has a horizontal asymptote with equation $y = 3$

On separate diagrams, sketch the curve with equation

(i) $y = f(x + 2)$

(3)

(ii) $y = -f(x)$

(3)

On each diagram, show clearly the coordinates of the points to which A and B are transformed and the equation of the asymptote.



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Question 1 continued

(Total for Question 1 is 6 marks)



2. The line l passes through the points $A(-3, 0)$ and $B\left(\frac{5}{2}, 22\right)$

- (a) Find the equation of l giving your answer in the form $y = mx + c$ where m and c are constants.

(3)

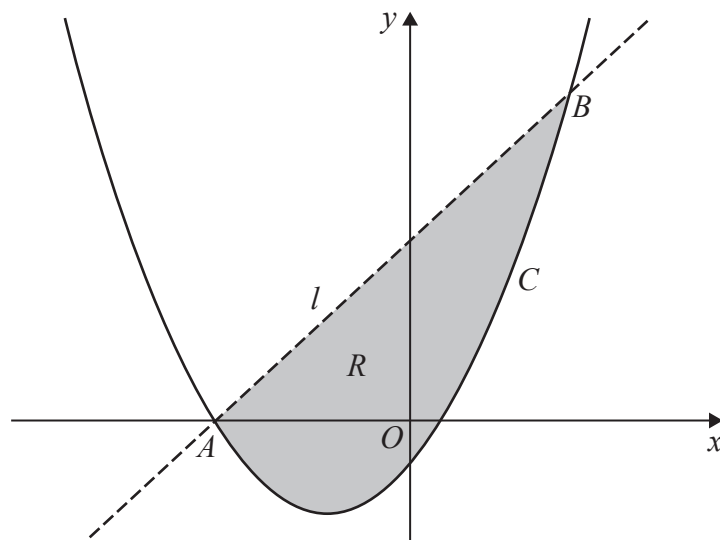


Figure 2

Figure 2 shows the line l and the curve C , which intersect at A and B .

Given that

- C has equation $y = 2x^2 + 5x - 3$
- the region R , shown shaded in Figure 2, is bounded by l and C

- (b) use inequalities to define R .

(2)

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 5 marks)



3. Relative to a fixed origin O

- the point A has coordinates $(3, -8)$
- the point B is such that $\vec{AB} = -5\mathbf{i} + 2\mathbf{j}$

(a) Find the exact value of $|\vec{OB}|$ (3)

(b) Find the size of angle OAB , giving your answer in degrees to one decimal place. (3)



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Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 6 marks)



4. The circle C has equation

$$x^2 + y^2 + 10x - 4y + 1 = 0$$

(a) Find

(i) the coordinates of the centre of C

(ii) the exact radius of C

(3)

The line with equation $y = k$, where k is a constant, cuts C at two distinct points.

(b) Find the range of values for k , giving your answer in set notation.

(3)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 6 marks)



5.

$$f(x) = \frac{x^5 - 12x^{\frac{1}{2}}}{4x}$$

(a) Write $f(x)$ in the form

$$ax^p + bx^q$$

where a , b , p and q are simplified constants.

(3)

(b) Hence find

$$\int f(x) \, dx$$

giving your answer in simplest form.

(3)



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Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 6 marks)



6.

$$f(x) = ax^3 + bx^2 + 18x + 9$$

where a and b are constants.

Given that $(x + 3)$ is a factor of $f(x)$

(a) show that

$$-3a + b = 5 \tag{2}$$

Given also that $f'(2) = 14$

(b) find the value of a and the value of b . (5)

[illegible]

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 7 marks)



7.

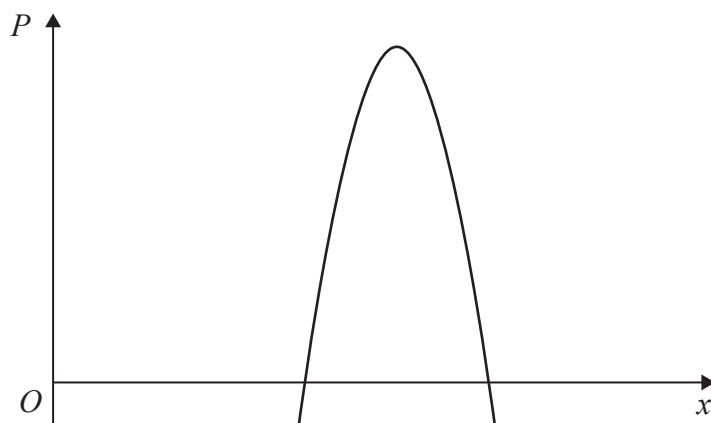


Figure 3

A company makes a particular type of chair.

The annual profit made by the company is modelled by the equation

$$P = -x^2 + 260x - 16450$$

where P is the profit measured in **thousands** of pounds and x is the selling price of the chair in pounds.

The graph of P against x is shown in Figure 3.

Using the model,

- (a) explain why £175 is not a sensible selling price for the chair. (2)

Given that the company made an annual profit of more than £200 000

- (b) find, according to the model, the highest possible selling price for the chair.

You must show your working clearly. (3)

- (c) Show that

$$P = a + b(x + c)^2$$

where a , b and c are constants to be found. (3)

The company wishes to maximise its annual profit.

State, according to the model,

- (d) (i) the maximum possible annual profit,
 (ii) the selling price of the chair that maximises the annual profit. (2)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 10 marks)



8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

Solve, for $0 \leq x \leq 90^\circ$, the equation

$$\sin^2 3x = 4 \cos^2 3x$$

giving your answers, in degrees, to one decimal place.

(5)



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Question 8 continued

Handwriting practice area with horizontal lines.

(Total for Question 8 is 5 marks)



9. An equation

$$x^2 - 3px + 5q + 4 = 0$$

where p and q are constants, has equal roots.

Show that

$$q = \frac{1}{20}(3p + 4)(3p - 4) \quad (3)$$



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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 3 marks)



10.

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

Solve the equation

$$2\log_3(x+1) = 1 + \log_3(x+7) \quad (5)$$

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 5 marks)



11.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

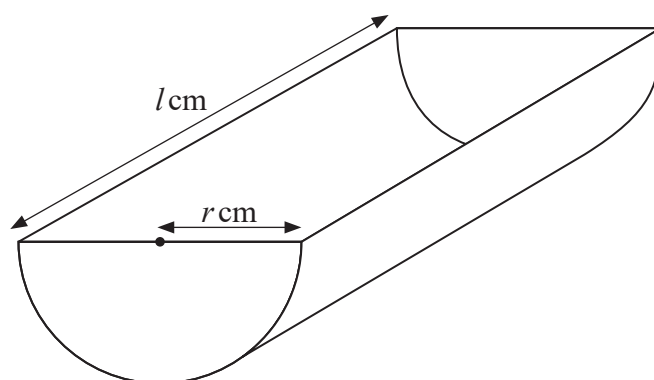


Figure 4

Figure 4 shows a design for a feeding trough.

The trough is modelled as a hollow, semicircular cylinder of radius r cm and length l cm.

The trough will be made from sheet metal of negligible thickness.

Given that the capacity of the trough will be $90\,000\pi$ cm³

- (a) show that the total area, A cm², of sheet metal required to make the trough is given by

$$A = \frac{180\,000\pi}{r} + \pi r^2 \quad (4)$$

- (b) Use calculus to find the radius of the trough for which A is a minimum. (4)

- (c) Show that the radius found in part (b) gives the minimum value of A . (2)

Given that the sheet metal costs £30 per square metre

- (d) calculate the minimum cost of sheet metal required to make one trough. (2)

- (e) State one assumption you have made in calculating your answer to part (d). (1)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 13 marks)



12.

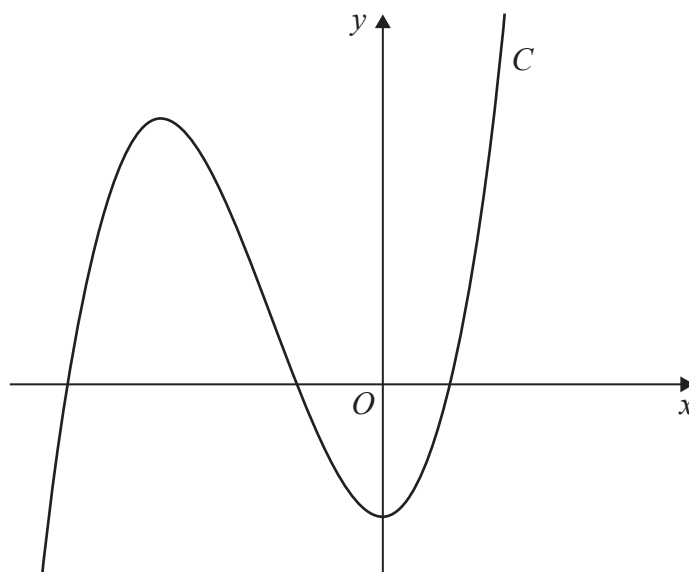


Figure 5

Figure 5 shows a sketch of the curve C .

The curve has a minimum turning point on the y -axis and a maximum turning point in quadrant 2.

- (a) On **Diagram 1**, sketch the graph of the gradient function for C .

[A copy of Figure 5 has been included on the next page to help you.]

(3)

The gradient function for C has **one** of the following equations

$$\frac{dy}{dx} = (x + 1)^2 - 2$$

$$\frac{dy}{dx} = 3x^2 + 9x$$

$$\frac{dy}{dx} = -x(x + 7)$$

$$\frac{dy}{dx} = x^2 - 5x$$

- (b) State the correct equation for the gradient function, fully justifying your answer.

(2)

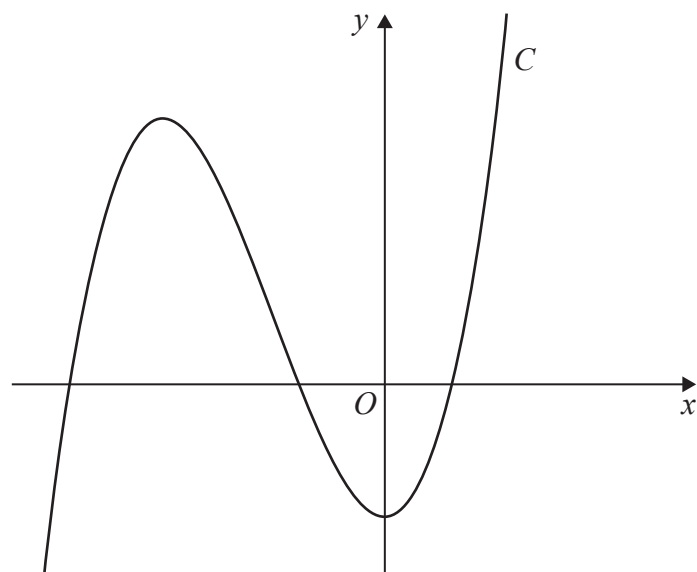


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Question 12 continued



Copy of Figure 5

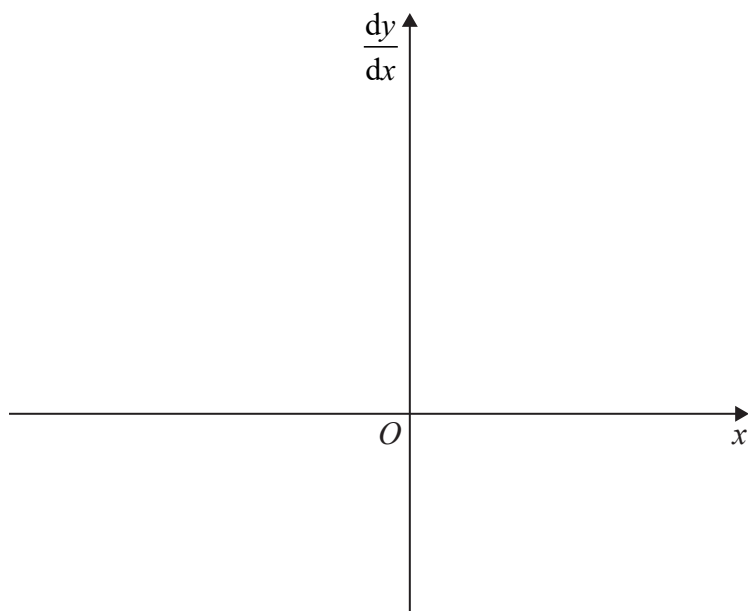


Diagram 1

(Total for Question 12 is 5 marks)



13.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The growth of a particular tree is monitored over a period of time.

The height, h metres, of this tree, t years after it was planted, is modelled by the equation

$$h = 31 - Ae^{-kt}$$

where A and k are positive constants.

Given that

- exactly 10 years after it was planted, the height of the tree was 6 m
 - exactly 20 years after it was planted, the height of the tree was 11 m
- (a) find a complete equation for h in terms of t , giving the value of each of A and k to 3 significant figures.

(4)

Use the equation of the model to answer parts (b), (c) and (d).

According to the model, there is a limit to the height to which this tree can grow.

- (b) Deduce this limit.

(1)

- (c) (i) Find the initial height of the tree.

- (ii) Hence explain whether this is a suitable model for the early growth of the tree.

(2)

- (d) (i) Find $\frac{dh}{dt}$, giving your answer in simplest form.

(2)

- (ii) Hence find the value of t for which the height of the tree is increasing at a rate of 30 cm per year.

(3)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Handwriting practice area with horizontal lines.

(Total for Question 13 is 12 marks)



14.

In this question you must show detailed reasoning.

$$f(x) = (1 + kx)^8$$

where k is a constant.

Given that the first 3 terms, in ascending powers of x , of the binomial series expansion of $f(x)$ are

$$1 + 8kx + px^2$$

where p is a constant,

(a) find p in terms of k .

(2)

$$g(x) = \left(a - \frac{2}{x}\right)f(x) \quad x \neq 0$$

where a is a constant.

Given that the first 3 terms, in ascending powers of x , of the series expansion of $g(x)$ are

$$-\frac{2}{x} - 21 - 90x$$

(b) find the possible pairs of values of a and k .

(5)

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Question 14 continued

Handwriting practice area with horizontal lines.



Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 7 marks)



15. $"n^3 + 4n$ is prime for $n \in \mathbb{N}"$ (I)

- (a) Determine whether statement (I) is always true, sometimes true or never true.
You must fully justify your answer.

(2)

$"n^3 + 5n$ is prime for $n \in \mathbb{N}"$ (II)

- (b) Determine whether statement (II) is always true, sometimes true or never true.
You must fully justify your answer.

(2)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Candidate Number

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Mock Paper Set 1

Time: 2 hours

Paper Reference **8MA0/01**

Mathematics

Advanced Subsidiary

Paper 1: Pure Mathematics

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 16 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. Find

$$\int \frac{2x^3 + 3}{x^2} dx$$

writing your answer in simplest form.

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 4 marks)



2. The triangle ABC is such that

- $AC = 12 \text{ cm}$
- $AB = 8 \text{ cm}$
- $\text{area } ABC = 30 \text{ cm}^2$

Find the two possible sizes of angle BAC , giving your answers in degrees to one decimal place.

(4)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3. By taking logarithms of both sides, solve the equation

$$2^{5k+3} = 3^{550}$$

giving the value of k to one decimal place.

(3)



DO NOT WRITE IN THIS AREA

Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 3 marks)



4. In the binomial expansion of

$$\left(k + \frac{1}{2}x\right)^9 \quad \text{where } k \text{ is a positive constant}$$

the coefficient of x^3 is 70

Find the value of k , giving your answer to 2 decimal places.

(3)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 3 marks)



5.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**(a) By letting $p = 2^x$, show that the equation

$$2 \times 4^x + 2^{x+3} = 1 + 2^{x-2}$$

can be rewritten in the form

$$8p^2 + 31p - 4 = 0 \quad (3)$$

(b) Hence solve

$$2 \times 4^x + 2^{x+3} = 1 + 2^{x-2} \quad (3)$$



DO NOT WRITE IN THIS AREA

Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



6. The points P , Q and R have coordinates $(-5, -5)$, $(7, 1)$ and $(9, -3)$ respectively.

(a) Show that angle $PQR = 90^\circ$

(3)

Given that P , Q and R each lie on a circle C

(b) find an equation for C

(3)



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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7.

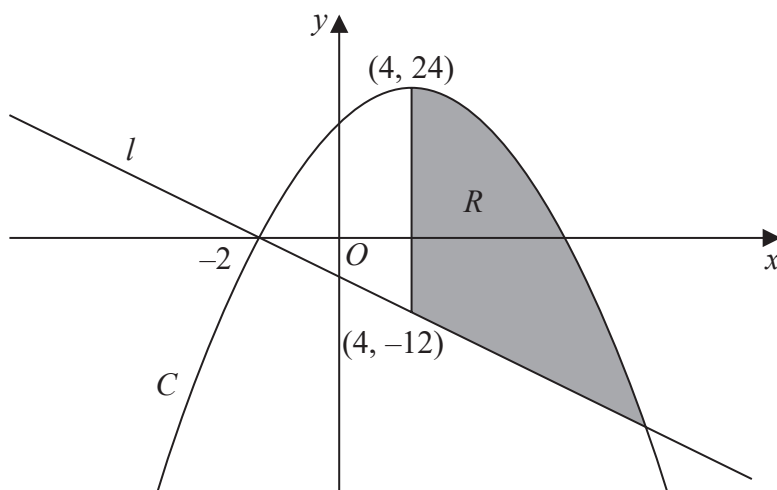


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = f(x)$ where

- $f(x)$ is a quadratic expression
- C has a maximum turning point at $(4, 24)$
- C cuts the negative x -axis at -2

(a) (i) Deduce the coordinates of the point at which C cuts the positive x -axis.

(ii) Hence, or otherwise, find $f(x)$.

(3)

The straight line l , also shown in Figure 1, passes through the points $(-2, 0)$ and $(4, -12)$

(b) Find an equation for l , writing your answer in the form $y = mx + c$, where m and c are constants to be found.

(3)

A shaded region R is shown in Figure 1.

(c) Use inequalities to define R .

(3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 9 marks)



8. A curve C has equation $y = g(x)$.

Given that

- $g'(x) = 6x^2 + 5x + k$ where k is a constant
- the graph of C cuts the y -axis at $(0, -10)$
- $(x + 4)$ is a factor of $g(x)$

find $g(x)$ in simplest form.

(6)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 6 marks)



DO NOT WRITE IN THIS AREA

Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



10.

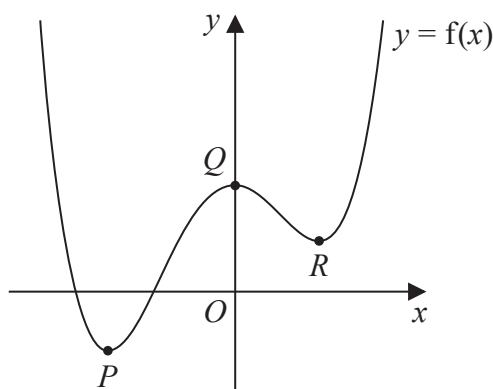


Figure 2

Figure 2 shows a sketch of the curve with equation $y = f(x)$.

The curve has exactly three stationary points

- a minimum turning point P at $(-3, -7)$
- a maximum turning point Q at $(0, 10)$
- a minimum turning point R at $(2, 5)$

(a) Deduce the values of x for which $f'(x) < 0$. Write your answer in set notation.

(2)

Given that the value of $f''(x)$ is -5 at one of the stationary points,

(b) explain why this must be point Q .

(1)

Given that the equation $f(x) = k$, where k is a constant, has **exactly** 3 roots,

(c) write down the two possible values for k .

(2)

On the next page there is an unlabelled copy of Figure 2 called Diagram 1.

(d) On Diagram 1 sketch a graph of the curve with equation $y = f'(x)$.

(3)



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Question 10 continued

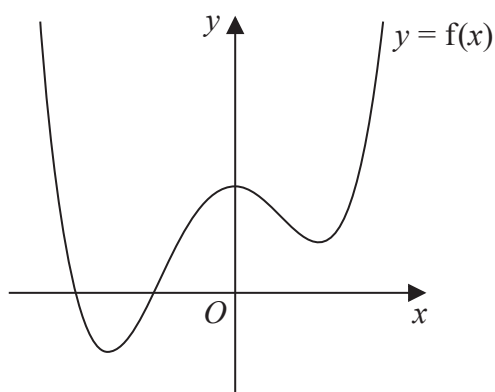


Diagram 1

(Total for Question 10 is 8 marks)



11. [In this question the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are due east and due north respectively.]

A boat travels in a straight line at constant speed.

Initially the boat has position $(-11\mathbf{i} - 2\mathbf{j})$ km relative to a fixed origin O

After 90 minutes the boat has position $(\mathbf{i} + 6\mathbf{j})$ km relative to O

- (a) Show that the speed of the boat is $p\sqrt{13} \text{ km h}^{-1}$, where p is a constant to be found. (3)

The boat continues in the same direction until it reaches point X

Given that X is due north east of O

- (b) find the position vector of X , making your method clear. (4)



DO NOT WRITE IN THIS AREA

Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 7 marks)



12. (i) A student states that

“ $n^2 - n + 5$ is a prime number for all $n \in \mathbb{N}$ ”

Show, by counter example, that the student's statement is **not** true.

(1)

(ii) Prove that

$$\tan \theta + \frac{1}{\tan \theta} \equiv \frac{1}{\sin \theta \cos \theta} \quad \theta \neq (90n)^\circ \quad n \in \mathbb{N}$$

(3)



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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 4 marks)



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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 7 marks)



14.

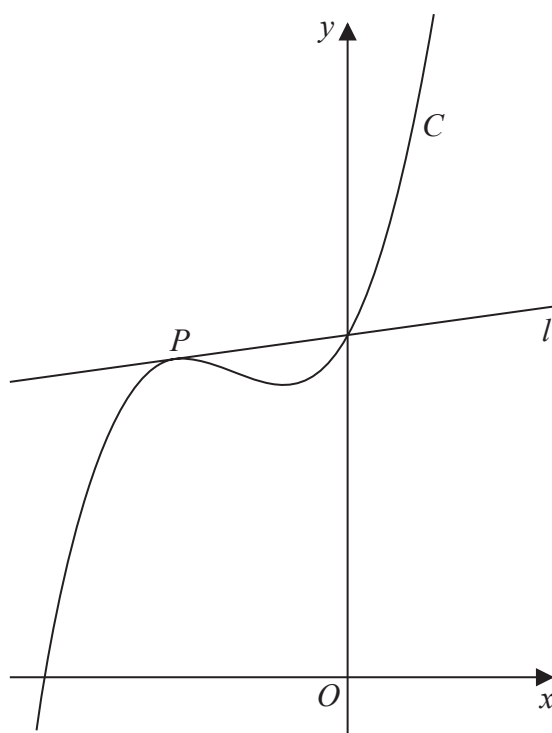


Figure 3

Figure 3 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (x + 4)(2x^2 + x + 7)$$

(a) Use the given information to state the values of x for which

$$f(x) > 0 \quad (1)$$

(b) Expand $(x + 4)(2x^2 + x + 7)$ writing your answer as a polynomial in simplest form.

(2)

The straight line l , also shown in Figure 3, has gradient k .

Given that

- l intersects C on the y -axis
- l is a tangent to C at a point P , as shown in Figure 3

(c) find the value of k .

(5)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 8 marks)



15.

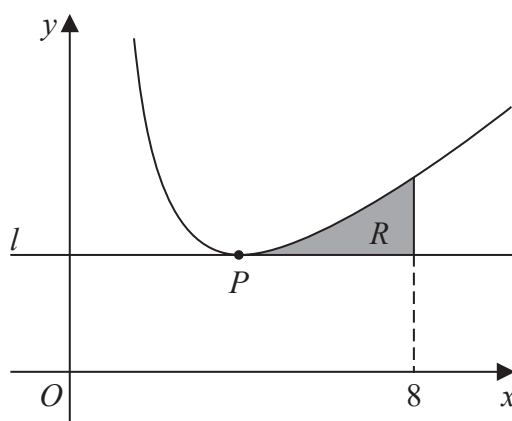


Figure 4

Figure 4 shows a sketch of the curve with equation

$$y = \frac{3x}{2} + \frac{32}{\sqrt{x^3}} \quad x > 0$$

The point P is the only stationary point on the curve.

(a) Show that the x coordinate of P is 4

(4)

The line l is parallel to the x -axis and passes through the point P

The region R , shown shaded in Figure 4, is bounded by the curve, the line l and the line with equation $x = 8$

(b) Use algebraic integration to find the exact area of R , writing your answer in the form $a + b\sqrt{2}$ where a and b are integers to be found.

(5)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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Question 15 continued

Lined area for writing the answer to Question 15.

(Total for Question 15 is 9 marks)



16.

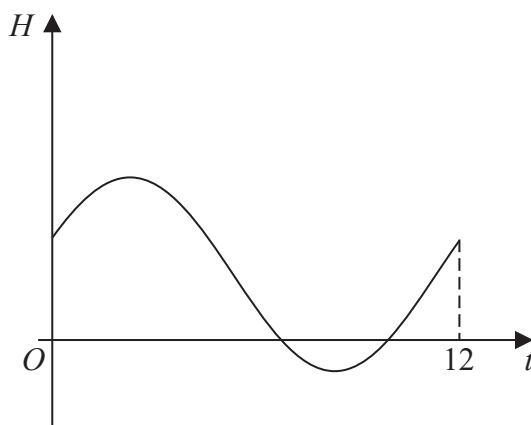


Figure 5

A horizontal path connects an island to the mainland.

On a particular morning, the height of the sea relative to the path, H m, is modelled by the equation

$$H = 0.8 + k \cos(30t - 70)^\circ$$

where k is a constant and t is number of hours after midnight.

Figure 5 shows a sketch of the graph of H against t .

Use the equation of the model to answer parts (a), (b) and (c).

- (a) Find the time of day at which the height of the sea is at its maximum.

(2)

Given that the maximum height of the sea relative to the path is 2 m,

- (b) (i) find a complete equation for the model,

- (ii) state the minimum height of the sea relative to the path.

(3)

It is safe to use the path when the sea is 10 centimetres or more **below** the path.

- (c) Find the times between which it is safe to use the path.

(Solutions relying entirely on calculator technology are not acceptable.)

(5)



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Question 16 continued

Lined area for writing the answer to Question 16.



Question 16 continued

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(Total for Question 16 is 10 marks)

(TOTAL FOR PAPER IS 100 MARKS)



Please check the examination details below before entering your candidate information

Candidate surname

Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Time 2 hours

Paper
reference

8MA0/01

Mathematics

Advanced Subsidiary

PAPER 1: Pure Mathematics

Mock Set 2

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. The line l_1 passes through the points $(6, -5)$ and $(2, 1)$

- (a) Find the equation of l_1 giving your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

The line l_2 is perpendicular to l_1 and passes through the origin.

- (b) Find the coordinates of the point of intersection of l_1 with l_2

(4)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2.

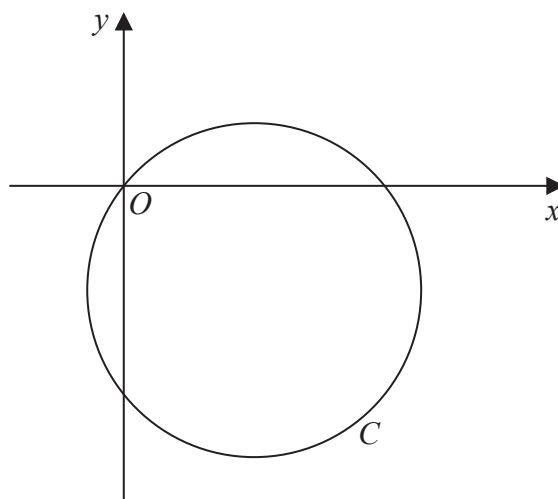


Figure 1

Figure 1 shows the circle C with equation

$$x^2 + y^2 - 12x + 10y = 0$$

(a) Find

(i) the coordinates of the centre of C

(ii) the exact radius of C

(3)

The straight line with equation $y = k$, where k is a positive constant, is a tangent to C at the point P

(b) Find the exact value of k

(1)

Given that C meets the y -axis at the origin O and at the point Q

(c) find the area of the triangle OPQ , making your method clear.

(3)



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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 7 marks)



3.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

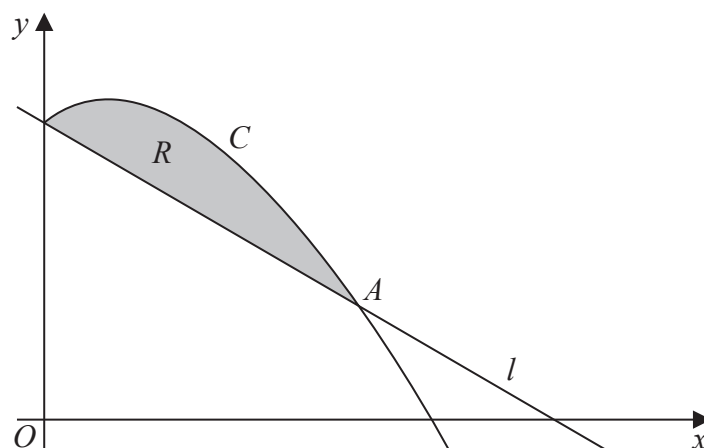


Figure 2

Figure 2 shows a sketch of part of the curve C with equation $y = 4x - 3x^{\frac{3}{2}} + 13$ and the line l with equation $y = 13 - 2x$

(a) Verify that the line and the curve intersect at the point $A(4, 5)$

(2)

The region R , shown shaded in Figure 2, is bounded by C and l

(b) Find the area of R

(4)



DO NOT WRITE IN THIS AREA

Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 6 marks)



4.

$$f(x) = x^4 - 2x^3 - 11x^2 + 12x + 36$$

- (a) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$ (2)

Given that $f(x) = (x - 3)^2(x + a)^2$, where a is a positive constant,

- (b) deduce the value of a (1)

- (c) Sketch the curve with equation $y = f(x)$, indicating clearly on your sketch the coordinates of the points at which the curve crosses or meets the axes. (4)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



5. The straight line l has equation

$$2x - y + 6 = 0$$

The curve C has equation

$$y = 2x^2 + kx + 9$$

where k is a constant.

Find the range of possible values of k for which l meets C at 2 distinct points.

Give your answer in set notation.

(5)



DO NOT WRITE IN THIS AREA

Question 5 continued

Handwriting practice area with horizontal lines.

(Total for Question 5 is 5 marks)



6. Given

$$3^x = 3\sqrt{3}(9^{y+1})$$

find an expression for y in terms of x in the form $y = ax + b$, where a and b are rational constants to be found.

(3)



DO NOT WRITE IN THIS AREA

Question 6 continued

Handwriting practice area with horizontal lines.

(Total for Question 6 is 3 marks)



7. Prove that, for all $x \in \mathbb{R}$,

$$4x^2 > 20x - 27$$

(3)

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 3 marks)



8. (a) Find the first four terms, in ascending powers of x , of the binomial expansion of

$$(2 + 3x)^6$$

simplifying each term.

(4)

- (b) Hence, without further calculation, write down the first four terms in the binomial expansion of

$$(2 - 3x)^6$$

(1)

- (c) Hence, find the first two non-zero terms in the binomial expansion of

$$\left[(2+3x)^6 + (2-3x)^6 \right]^2$$

(2)

- (d) Find the term that is independent of x in the binomial expansion of

$$\left[(2+ax)^n + (2-ax)^n \right]^p$$

giving your answer in the form 2^y , where y is in terms of n and p

(2)



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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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Question 8 continued

Lined area for writing answers.

(Total for Question 8 is 9 marks)



9.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$5 \cos \theta = 24 \tan \theta$$

can be written as

$$5 \sin^2 \theta + 24 \sin \theta - 5 = 0 \quad (4)$$

(b) Hence solve, for $-180^\circ \leq x < 540^\circ$

$$5 \cos x = 24 \tan x$$

giving your answers to one decimal place. (3)

(c) Deduce the number of solutions in the interval $-180^\circ \leq x < 540^\circ$ of the equation

$$5 \cos(8x) = 24 \tan(8x) \quad (1)$$



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Question 9 continued

Lined area for writing the answer to Question 9.



Question 9 continued

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 8 marks)



10.

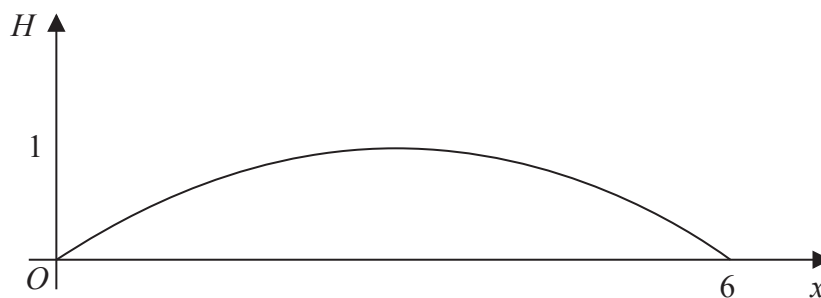


Figure 3

Figure 3 is a graph showing the trajectory of a snowboarder during a jump.

The vertical height, H metres, of the snowboarder above the ground has been plotted against the horizontal distance travelled, x metres, measured from where the snowboarder took off.

The snowboarder is modelled as a particle travelling in a vertical plane above horizontal ground.

The snowboarder

- reaches a maximum height of one metre above the ground
- lands on the ground at a point 6 metres from where the snowboarder took off

Given that H is modelled as a **quadratic** function in x

(a) find H in terms of x

(3)

The snowboarder passes over two vertical poles of height 0.5 metres. The poles are in the same plane as the snowboarder's motion.

(b) Use the model to find the greatest distance between the two poles.

(4)

(c) Give one limitation of the model.

(1)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



11.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**Find any real values of x such that

$$2\log_6(x+3) = 2 - \log_6(4-x)$$

(7)



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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 7 marks)



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Question 12 continued

Lined area for writing the answer to Question 12.



Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 10 marks)



13.

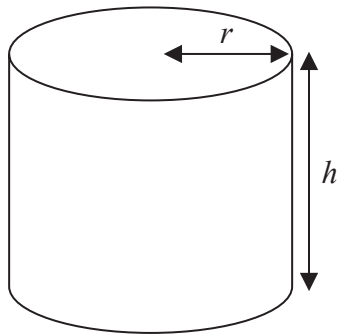


Figure 4

Figure 4 shows a closed cylindrical can of radius r cm and height h cm.

Given that the volume of the can is 400 cm^3

(a) show that the total surface area of the can, $A \text{ cm}^2$, is given by

$$A = 2\pi r^2 + \frac{800}{r} \quad (4)$$

(b) Find the exact radius of the can for which A is a minimum. (4)

(c) By finding $\frac{d^2A}{dr^2}$, show that the radius found in part (b) gives the minimum value of A

(2)

(d) Calculate the minimum value of A (2)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Handwriting practice area with horizontal lines.

(Total for Question 13 is 12 marks)



14. In the triangle OAB

- $\vec{OA} = 3\mathbf{i} - 4\mathbf{j}$
- $|\vec{OB}| = 10$
- angle $AOB = 60^\circ$

(a) Show that $|\vec{AB}| = 5\sqrt{3}$

(4)

Given further that $\vec{BA} = 4\sqrt{3}\mathbf{i} + p\mathbf{j}$, where p is a positive constant,

(b) find $\vec{OB}$, giving your answer using $\mathbf{i}, \mathbf{j}$ notation.

(4)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 8 marks)

TOTAL FOR PAPER IS 100 MARKS



Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Mock Paper Set 3

Time: 2 hours

Paper reference **8MA0/01**

Mathematics
Advanced Subsidiary
PAPER 1: Pure Mathematics

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. The curve C has equation

$$y = \frac{3}{2}x^3 - 5x - \frac{10}{x}$$

(a) Find $\frac{dy}{dx}$ giving your answer in simplest form.

(3)

Given that the point $P(-2, 3)$ lies on C ,

(b) find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(4)



DO NOT WRITE IN THIS AREA

Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 7 marks)



2.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Use algebra to solve the inequality

$$3x^2 > 7x + 10$$

Write your answer in set notation.

(3)



DO NOT WRITE IN THIS AREA

Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 3 marks)



3.

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 6x^3 - (2a + 5)x^2 + 21x + a$$

where a is a constant.

Given that $(x - 2)$ is a factor of $f(x)$

(a) show that $a = 10$ (2)

(b) Hence solve the equation

$$f(x) = 0 \quad (4)$$

(c) Write down the largest solution of $f(-x) = 0$ (1)



DO NOT WRITE IN THIS AREA

Question 3 continued

Handwriting practice area with horizontal lines.

(Total for Question 3 is 7 marks)



4. Given that $\vec{PQ} = -3\mathbf{i} + 12\mathbf{j}$

(a) find $|\vec{PQ}|$, giving your answer in simplest form.

(2)

Given also that relative to a fixed origin O

- the point P has position vector $9a\mathbf{i} - 4b\mathbf{j}$
- the point Q has position vector $6b\mathbf{i} + 15a\mathbf{j}$

where a and b are constants,

(b) find the value of a and the value of b .

(3)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 5 marks)



5.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(i) Solve the equation

$$3(x - 2\sqrt{5}) = x\sqrt{5}$$

giving your answer in simplest form.

(3)

(ii) Find the exact value of x for which

$$e^{4x-1} = 5e^{\frac{1}{2}x}$$

(3)



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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 6 marks)



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Question 6 continued

Lined area for writing the answer to Question 6.



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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 7 marks)



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7. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{3}{4}x\right)^7$$

giving each term in simplest form.

(4)

Given that

$$f(x) = (x + 5)\left(2 - \frac{3}{4}x\right)^7$$

- (b) find the coefficient of x^2 in the expansion of $f(x)$

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 6 marks)



8.

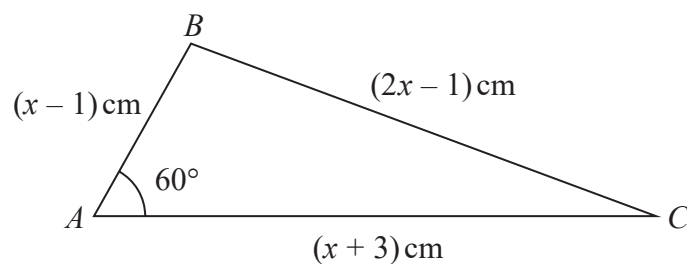


Figure 1

Figure 1 shows a sketch of triangle ABC .

Given that

- $AB = (x - 1) \text{ cm}$
- $BC = (2x - 1) \text{ cm}$
- $AC = (x + 3) \text{ cm}$
- $\text{angle } BAC = 60^\circ$

(a) show that $x^2 - 2x - 4 = 0$

(3)

(b) Hence find the area of triangle ABC giving your answer to 3 significant figures.

(3)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 6 marks)



9.

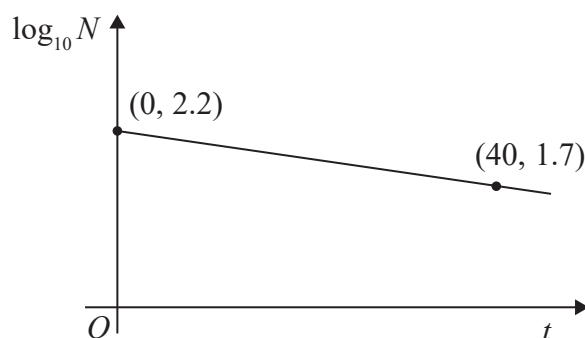


Figure 2

The number of fish in a lake is being monitored.

The line shown in Figure 2 models the linear relationship between $\log_{10} N$ and t , where

- N is the number of fish in thousands
- t is the number of years after monitoring began

The line passes through the points $(0, 2.2)$ and $(40, 1.7)$

Using this information,

- (a) find an equation for this line, (2)
- (b) find a complete equation for the model in the form

$$N = ab^t$$

where a and b are constants.

Give the value of a and the value of b , each to 3 significant figures. (3)

With reference to the model interpret,

- (c) (i) the value of a
- (ii) the value of b (2)
- (d) Find, according to the model, the number of fish when $t = 10$, giving your answer to the nearest 1000 (1)

The model predicts that T years after monitoring began, the number of fish will fall below 20 000 for the first time.

- (e) Find the value of T , giving your answer to the nearest integer. (2)
- (Solutions relying entirely on calculator technology are not acceptable.)*
- (f) Give a reason why the model may not be realistic. (1)



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Lined area for writing.



Question 9 continued

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Question 9 continued

Lined area for writing answers.

(Total for Question 9 is 11 marks)



10.

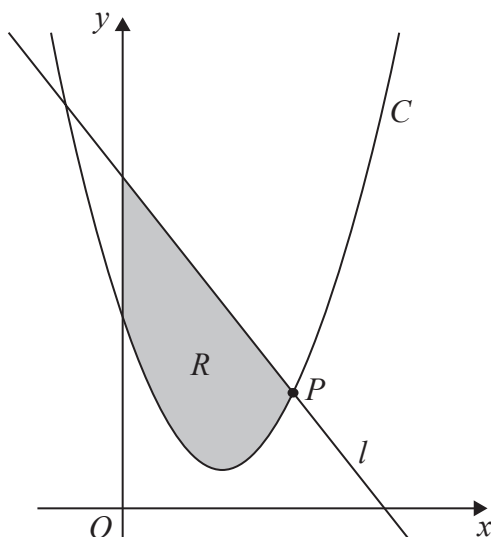


Figure 3

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of the curve C with equation

$$y = 2x^2 - 7x + 8$$

and a sketch of the line l with equation

$$y = -3x + 14$$

The line l intersects the curve C at the point P as shown in Figure 3.

- (a) Find the x coordinate of P (3)

The region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

- (b) Use algebraic integration to find the area of R (4)



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Question 10 continued

Lined area for writing the answer to Question 10.



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Question 10 continued

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 7 marks)



11.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The furthest distance a particular electric car can travel at a constant speed, starting from a full charge, is modelled by the equation

$$y = \frac{1}{250}x^3 - \frac{3}{5}x^2 + 24x \quad 0 \leq x \leq 70$$

where y is the furthest distance in miles and x is the constant speed in miles per hour.

Use the model to answer parts (a), (b), (c) and (d).

- (a) Find the furthest distance the car can travel at a constant speed of 50 miles per hour, starting from a full charge. (1)
- (b) Use calculus to find the value of x for which y is a maximum, giving your answer to 3 significant figures. (4)
- (c) Using $\frac{d^2y}{dx^2}$ prove that y is maximised for the value of x found in part (b). (2)
- (d) Hence find the maximum distance the car can travel at a constant speed, starting from a full charge, giving your answer to the nearest mile. (2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Handwriting practice area with horizontal lines.

(Total for Question 11 is 9 marks)



12.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Use the laws of logarithms to solve

$$4(\log_x 3 + \log_6 \sqrt{3}) = 3 \log_x 3 - \log_6 4$$

giving your answer as a simplified surd.

(5)



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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 5 marks)



13.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Prove that

$$\frac{\tan \theta (16 + 9 \sin^2 \theta)}{3 \sin \theta + 5 \tan \theta} \equiv 5 - 3 \cos \theta \quad \theta \neq (90n)^\circ \quad n \in \mathbb{Z} \quad (4)$$

(b) Hence solve for $0 < x < 180^\circ$

$$\frac{\tan 2x (16 + 9 \sin^2 2x)}{3 \sin 2x + 5 \tan 2x} = \frac{4}{\cos 2x} - 3$$

giving your answers to one decimal place.

(4)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Handwriting practice area with horizontal lines.

(Total for Question 13 is 8 marks)



14. (a) Find

$$\int \frac{5x(5x-6)-9}{2\sqrt{x}} dx$$

giving each term in simplest form.

(4)

Given that

$$\int_1^k \frac{5x(5x-6)-9}{2\sqrt{x}} dx < 6\sqrt{k} + 14$$

where k is a constant, $k > 2$

(b) (i) show that

$$k^2 - 2k - 3 < 0$$

(ii) Hence find the range of values of k

(5)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Handwriting practice area with horizontal lines.

(Total for Question 14 is 9 marks)



15. Prove, using algebra, that for all $x \in \mathbb{R}$

$$(x + 7)(x + 1) > 2x - 3$$

(4)

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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS



Pearson Edexcel Level 3 GCE

Mathematics

Advanced Subsidiary

Paper 1: Pure Mathematics

Sample assessment material for first teaching
September 2017
Time: 2 hours

Paper Reference(s)

8MA0/01

You must have:

Mathematical Formulae and Statistical Tables
Calculator

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** the questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
- *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 17 questions in this question paper. The total mark for this paper is 100.
- The marks for each question are shown in brackets
- *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- If you change your mind about an answer, cross it out and put your new answer and any working underneath.

Answer ALL questions. Write your answers in the spaces provided.

- 1.** The line l passes through the points $A(3, 1)$ and $B(4, -2)$.

Find an equation for l .

(3)

(Total for Question 1 is 3 marks)

2. The curve C has equation

$$y = 2x^2 - 12x + 16$$

Find the gradient of the curve at the point $P(5, 6)$.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(4)

(Total for Question 2 is 4 marks)

3. Given that the point A has position vector $3\mathbf{i} - 7\mathbf{j}$ and the point B has position vector $8\mathbf{i} + 3\mathbf{j}$,

(a) find the vector $\overrightarrow{AB}$.

(2)

(b) Find $|\overrightarrow{AB}|$. Give your answer as a simplified surd.

(2)

(Total for Question 3 is 4 marks)

4.

$$f(x) = 4x^3 - 12x^2 + 2x - 6$$

(a) Use the factor theorem to show that $(x - 3)$ is a factor of $f(x)$.

(2)

(b) Hence show that 3 is the only real root of the equation $f(x) = 0$

(4)

(Total for Question 4 is 6 marks)

5. Given that

$$f(x) = 2x + 3 + \frac{12}{x^2}, \quad x > 0$$

show that $\int_1^{2\sqrt{2}} f(x) dx = 16 + 3\sqrt{2}$

(5)

(Total for Question 5 is 5 marks)

6. Prove, from first principles, that the derivative of $3x^2$ is $6x$.

(4)

(Total for Question 6 is 4 marks)

7. (a) Find the first 3 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{x}{2}\right)^7, \text{ giving each term in its simplest form.}$$

(4)

(b) Explain how you would use your expansion to give an estimate for the value of 1.995^7

(1)

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Question 7 continued

(Total for Question 7 is 5 marks)

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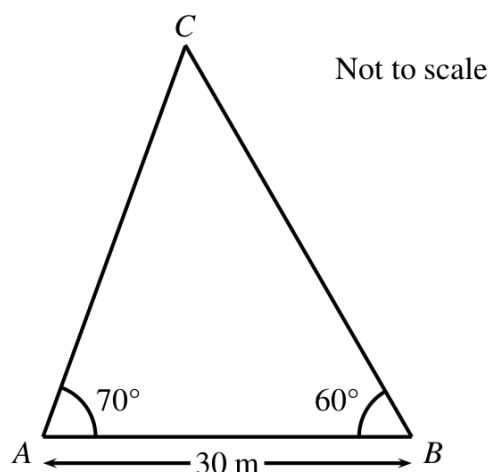


Figure 1

A triangular lawn is modelled by the triangle ABC , shown in Figure 1. The length AB is to be 30 m long.

Given that angle $BAC = 70^\circ$ and angle $ABC = 60^\circ$,

(a) calculate the area of the lawn to 3 significant figures.

(4)

(b) Why is your answer unlikely to be accurate to the nearest square metre?

(1)

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[illegible]

9. Solve, for $360^\circ \leq x < 540^\circ$,

$$12\sin^2 x + 7\cos x - 13 = 0$$

Give your answers to one decimal place.

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(5)

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[illegible]

11. (a) Prove that for all positive values of x and y

$$\sqrt{xy} \leq \frac{x+y}{2}$$

(2)

(b) Prove by counter example that this is not true when x and y are both negative.

(1)

(Total for Question 11 is 3 marks)

12. A student was asked to give the exact solution to the equation

$$2^{2x+4} - 9(2^x) = 0$$

The student's attempt is shown below:

$$2^{2x+4} - 9(2^x) = 0$$

$$2^{2x} + 2^4 - 9(2^x) = 0$$

$$\text{Let } 2^x = y$$

$$y^2 - 9y + 8 = 0$$

$$(y - 8)(y - 1) = 0$$

$$y = 8 \text{ or } y = 1$$

$$\text{So } x = 3 \text{ or } x = 0$$

(a) Identify the two errors made by the student.

(2)

(b) Find the exact solution to the equation.

(2)

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13. (a) Factorise completely $x^3 + 10x^2 + 25x$

(2)

(b) Sketch the curve with equation

$$y = x^3 + 10x^2 + 25x$$

showing the coordinates of the points at which the curve cuts or touches the x -axis.

(2)

The point with coordinates $(-3, 0)$ lies on the curve with equation

$$y = (x + a)^3 + 10(x + a)^2 + 25(x + a)$$

where a is a constant.

(c) Find the two possible values of a .

(3)

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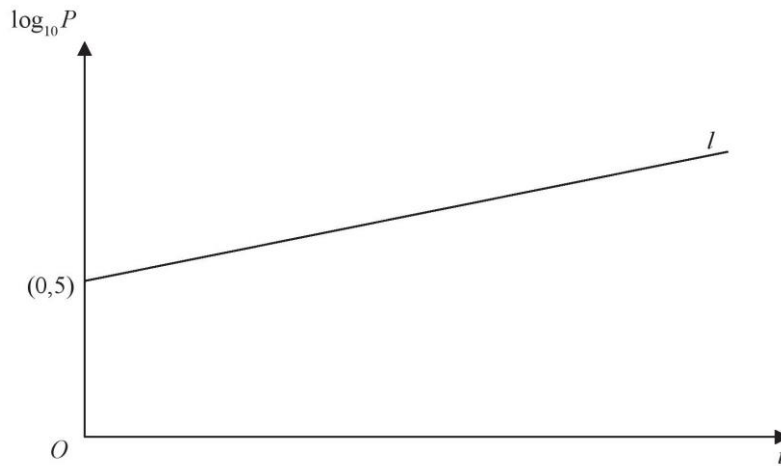


Figure 2

A town's population, P , is modelled by the equation $P = ab^t$, where a and b are constants and t is the number of years since the population was first recorded.

The line l shown in Figure 2 illustrates the linear relationship between t and $\log_{10} P$ for the population over a period of 100 years.

The line l meets the vertical axis at $(0, 5)$ as shown. The gradient of l is $\frac{1}{200}$.

- (a) Write down an equation for l . (2)
- (b) Find the value of a and the value of b . (4)
- (c) With reference to the model interpret
 - (i) the value of the constant a ,
 - (ii) the value of the constant b (2)
- (d) Find
 - (i) the population predicted by the model when $t = 100$, giving your answer to the nearest hundred thousand,
 - (ii) the number of years it takes the population to reach 200 000, according to the model. (3)
- (e) State two reasons why this may not be a realistic population model. (2)

[illegible]

(Total for Question 1 is 15 marks)

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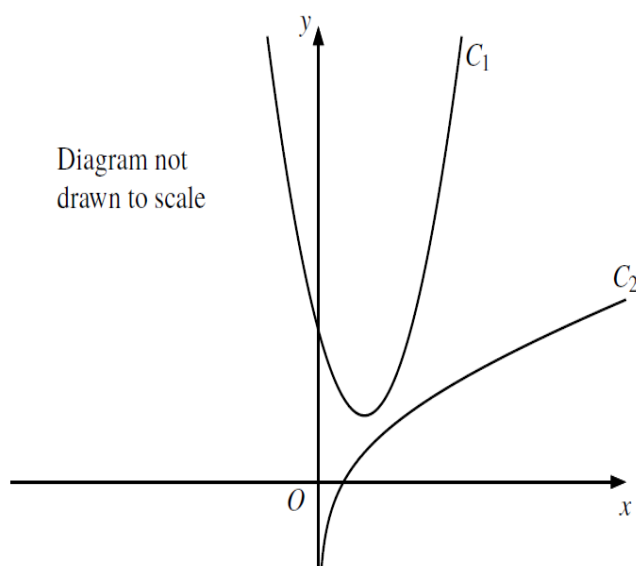


Figure 3

The curve C_1 , shown in Figure 3, has equation $y = 4x^2 - 6x + 4$.

The point $P\left(\frac{1}{2}, 2\right)$ lies on C_1

The curve C_2 , also shown in Figure 3, has equation $y = \frac{1}{2}x + \ln(2x)$.

The normal to C_1 at the point P meets C_2 at the point Q .

Find the exact coordinates of Q .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(8)

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[illegible]

(Total for Question 15 is 6 marks)

16.

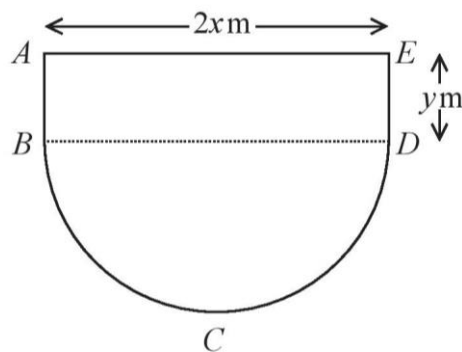


Figure 4

Figure 4 shows the plan view of the design for a swimming pool.

The shape of this pool $ABCDEA$ consists of a rectangular section $ABDE$ joined to a semicircular section BCD as shown in Figure 4.

Given that $AE = 2x$ metres, $ED = y$ metres and the area of the pool is 250 m^2 ,

(a) show that the perimeter, P metres, of the pool is given by

$$P = 2x + \frac{250}{x} + \frac{\pi x}{2} \quad (4)$$

(b) Explain why $0 < x < \sqrt{\frac{500}{\pi}}$ (2)

(c) Find the minimum perimeter of the pool, giving your answer to 3 significant figures. (4)

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[illegible]

(Total for Question 10 is 10 marks)

17. A circle C with centre at $(-2, 6)$ passes through the point $(10, 11)$.

(a) Show that the circle C also passes through the point $(10, 1)$.

(3)

The tangent to the circle C at the point $(10, 11)$ meets the y axis at the point P
and the tangent to the circle C at the point $(10, 1)$ meets the y axis at the point Q .

(b) Show that the distance PQ is 58 explaining your method clearly.

(7)

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TOTAL FOR PAPER IS 100 MARKS

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Surname

Other names

Pearson Edexcel
Level 3 GCE

Centre Number

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Mathematics

Advanced Subsidiary
Paper 1: Pure Mathematics

Specimen Paper

Time: 2 hours

Paper Reference

8MA0/01

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator allowed by the regulations of the Joint Council for Qualifications. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 14 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Pearson

Answer ALL questions. Write your answers in the spaces provided.

1. A curve has equation

$$y = 2x^3 - 2x^2 - 2x + 8$$

- (a) Find $\frac{dy}{dx}$

(2)

- (b) Hence find the range of values of x for which y is increasing.
Write your answer in set notation.

(4)

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. The quadrilateral $OABC$ has $\vec{OA} = 4\mathbf{i} + 2\mathbf{j}$, $\vec{OB} = 6\mathbf{i} - 3\mathbf{j}$ and $\vec{OC} = 8\mathbf{i} - 20\mathbf{j}$.

(a) Find $\vec{AB}$.

(2)

(b) Show that quadrilateral $OABC$ is a trapezium.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



- 3.** A tank, which contained water, started to leak from a hole in its base.

The volume of water in the tank 24 minutes after the leak started was 4 m^3

The volume of water in the tank 60 minutes after the leak started was 2.8 m^3

The volume of water, $V \text{ m}^3$, in the tank t minutes after the leak started, can be described by a linear model between V and t .

- (a) Find an equation linking V with t .

(4)

Use this model to find

- (b) (i) the initial volume of water in the tank,

- (ii) the time taken for the tank to empty.

(3)

- (c) Suggest a reason why this linear model may not be suitable.

(1)

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 8 marks)



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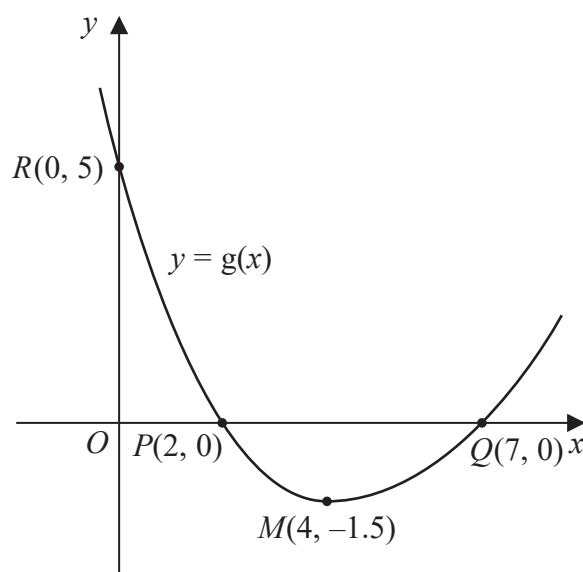


Figure 1

Figure 1 shows a sketch of the curve with equation $y = g(x)$.

The curve has a single turning point, a minimum, at the point $M(4, -1.5)$.

The curve crosses the x -axis at two points, $P(2, 0)$ and $Q(7, 0)$.

The curve crosses the y -axis at a single point $R(0, 5)$.

(a) State the coordinates of the turning point on the curve with equation $y = 2g(x)$. (1)

(b) State the largest root of the equation

$$g(x + 1) = 0 \quad (1)$$

(c) State the range of values of x for which $g'(x) \leq 0$ (1)

Given that the equation $g(x) + k = 0$, where k is a constant, has no real roots,

(d) state the range of possible values for k . (1)



DO NOT WRITE IN THIS AREA

Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 4 marks)



5.

$$f(x) = x^3 + 3x^2 - 4x - 12$$

- (a) Using the factor theorem, explain why $f(x)$ is divisible by $(x + 3)$. (2)

- (b) Hence fully factorise $f(x)$. (3)

- (c) Show that $\frac{x^3 + 3x^2 - 4x - 12}{x^3 + 5x^2 + 6x}$ can be written in the form $A + \frac{B}{x}$ where A and B are integers to be found.



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 8 marks)



6. (i) Use a counter example to show that the following statement is false.

“ $n^2 - n - 1$ is a prime number, for $3 \leq n \leq 10$.”

(2)

- (ii) Prove that the following statement is always true.

“The difference between the cube and the square of an odd number is even.”

For example $5^3 - 5^2 = 100$ is even.

(4)



DO NOT WRITE IN THIS AREA

Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 6 marks)



7. (a) Expand $\left(1 + \frac{3}{x}\right)^2$ simplifying each term.

(2)

(b) Use the binomial expansion to find, in ascending powers of x , the first four terms in the expansion of

$$\left(1 + \frac{3}{4}x\right)^6$$

simplifying each term.

(4)

(c) Hence find the coefficient of x in the expansion of

$$\left(1 + \frac{3}{x}\right)^2 \left(1 + \frac{3}{4}x\right)^6$$

(2)



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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 8 marks)



8.

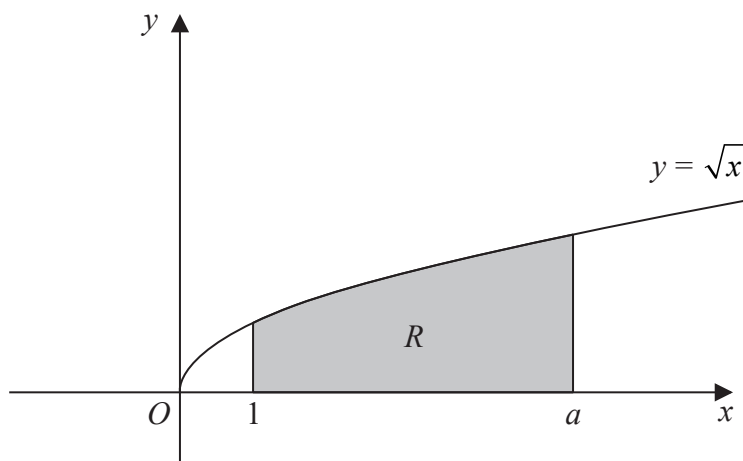


Figure 2

Figure 2 shows a sketch of the curve with equation $y = \sqrt{x}$, $x \geq 0$

The region R , shown shaded in Figure 2, is bounded by the curve, the line with equation $x = 1$, the x -axis and the line with equation $x = a$, where a is a constant.

Given that the area of R is 10

(a) find, in simplest form, the value of

(i) $\int_1^a \sqrt{8x} \, dx$

(ii) $\int_0^a \sqrt{x} \, dx$

(4)

(b) show that $a = 2^k$, where k is a rational constant to be found.

(4)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 8 marks)



9. Find any real values of x such that

$$2 \log_4(2 - x) - \log_4(x + 5) = 1$$

(6)

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Question 9 continued

Lined area for writing the answer to Question 9.

(Total for Question 9 is 6 marks)



10. A circle C has centre $(2, 5)$. Given that the point $P(-2, 3)$ lies on C .

(a) find an equation for C .

(3)

The line l is the tangent to C at the point P . The point $Q(2, k)$ lies on l .

(b) Find the value of k .

(5)



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Question 10 continued

Lined area for writing the answer to Question 10.

(Total for Question 10 is 8 marks)



- 11. (i)** Solve, for $-90^\circ \leq \theta < 270^\circ$, the equation,

$$\sin(2\theta + 10^\circ) = -0.6$$

giving your answers to one decimal place.

(5)

- (ii) (a) A student's attempt at the question

“Solve, for $-90^\circ < x < 90^\circ$, the equation $7 \tan x = 8 \sin x$ ”

is set out below.

$$7 \tan x = 8 \sin x$$

$$7 \times \frac{\sin x}{\cos x} = 8 \sin x$$

$$7 \sin x = 8 \sin x \cos x$$

$$7 = 8 \cos x$$

$$\cos x = \frac{7}{8}$$

$$x = 29.0^\circ \text{ (to 3 sf)}$$

Identify two mistakes made by this student, giving a brief explanation of each mistake.

(2)

- (b) Find the smallest positive solution to the equation

$$7 \tan(4\alpha + 199^\circ) = 8 \sin(4\alpha + 199^\circ)$$

(2)

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Question 11 continued

Lined area for writing the answer to Question 11.

(Total for Question 11 is 9 marks)



12.

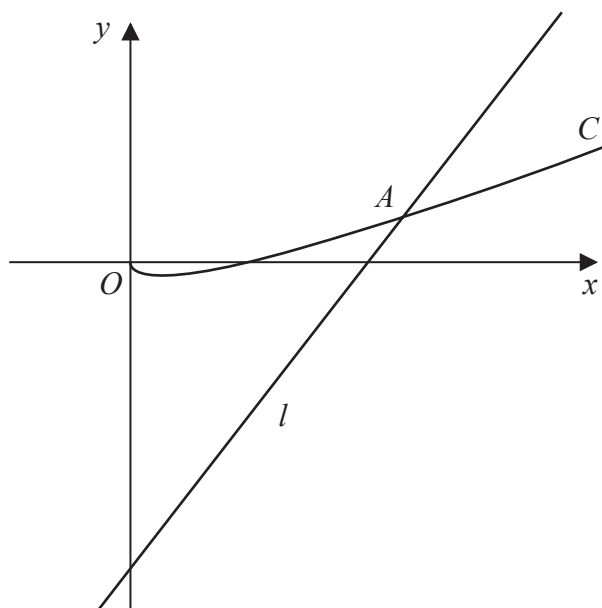


Figure 3

Figure 3 shows a sketch of the curve C with equation $y = 3x - 2\sqrt{x}$, $x \geq 0$ and the line l with equation $y = 8x - 16$

The line cuts the curve at point A as shown in Figure 3.

(a) Using algebra, find the x coordinate of point A .

(5)

(b)

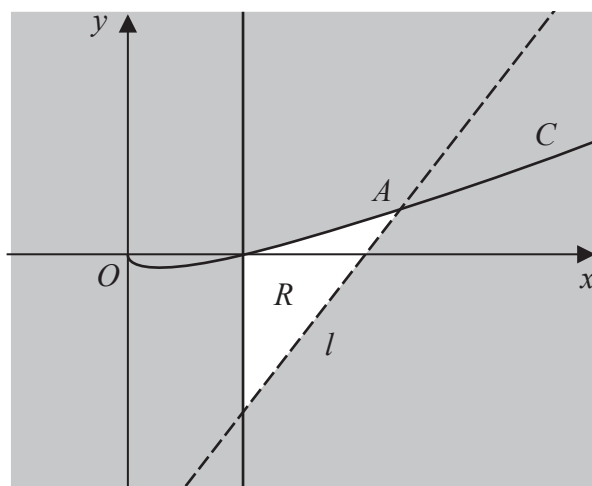


Figure 4

The region R is shown unshaded in Figure 4. Identify the inequalities that define R .

(3)



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Question 12 continued

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Question 12 continued

Lined area for writing the answer to Question 12.

(Total for Question 12 is 8 marks)



13. The growth of pond weed on the surface of a pond is being investigated.

The surface area of the pond covered by the weed, $A \text{ m}^2$, can be modelled by the equation

$$A = 0.2e^{0.3t}$$

where t is the number of days after the start of the investigation.

(a) State the surface area of the pond covered by the weed at the start of the investigation. **(1)**

(b) Find the rate of increase of the surface area of the pond covered by the weed, in m^2/day , exactly 5 days after the start of the investigation. (2)

Given that the pond has a surface area of 100m^2 ,

(c) find, to the nearest hour, the time taken, according to the model, for the surface of the pond to be fully covered by the weed. (4)

The pond is observed for one month and by the end of the month 90% of the surface area of the pond was covered by the weed.

(d) Evaluate the model in light of this information, giving a reason for your answer. (1)

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Question 13 continued

Lined area for writing the answer to Question 13.

(Total for Question 13 is 8 marks)



14.

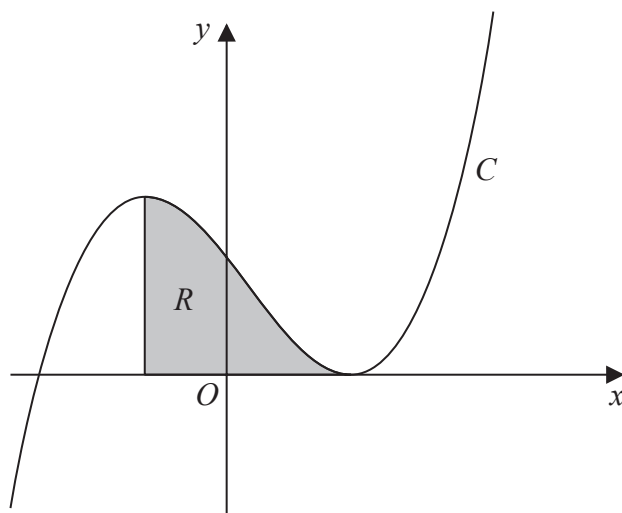


Figure 5

Figure 5 shows a sketch of the curve C with equation $y = (x - 2)^2(x + 3)$

The region R , shown shaded in Figure 5, is bounded by C , the vertical line passing through the maximum turning point of C and the x -axis.

Find the exact area of R .

(Solutions based entirely on graphical or numerical methods are not acceptable.)

(9)

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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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(Total for Question 14 is 9 marks)

TOTAL FOR PAPER IS 100 MARKS

